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Characters' Networks in Fanfiction: Exploring Characters Adaptations and Evolution through Networks.

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Abstract

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This study explores the reconstruction of character relationships and interactions in fanfiction using data from 97 fanfictions based on the movie "The Rocky Horror Picture Show" from the Ao3 platform.

The study employed a mixed-method approach, utilizing quantitative analysis to measure the character interaction relationships in the texts. It mainly used network analysis combined with image clustering tools to identify common character combinations, important central characters, and changes in character relationships across different stories. This approach helped transform stories of varying lengths into intuitive network structures, allowing for the analysis of significant character relationships from different perspectives. Finally, the analysis results were supplemented with close reading method. The findings demonstrate that this study, by integrating network analysis and visual analysis with close reading, fosters a new collaborative practice between close reading and distant reading.

The results show that fanfiction tends to preserve key character relationships from the original work. Moreover, within the fanfiction networks, more secondary characters gained prominence, altering the original character pairings and relationship settings. Preliminary character network results based on verb types suggest that fan authors may focus more on action descriptions rather than on explicitly describing dialogue occurrences.

Character co-occurrence networks from fanfictions were constructed using three different methods based on varying settings of character interaction information. Each method focused on different aspects of character co-occurrence. Despite variations in the selection criteria for fanfictions and the methods for constructing character networks according to specific research objectives, the analysis results are consistent with some established findings in fanfiction studies. This study also raises reflections on using computational tools to abstract story texts into networks. The abstraction of texts into networks inevitably leads to some loss of information, and through reflection on the methods of network construction and the data analysis process, the discussion on quantitative research methods is further enriched.

Also, it is important to acknowledge the limitations of this study, and future research is expected to further enhance and expand these aspects.

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Preface

First and foremost, I would like to express my heartfelt gratitude to my supervisor, Federico Pianzola. The process of completing this thesis has been far from smooth sailing, and I am deeply thankful for his patient guidance through the challenges I encountered. His willingness to trust me, helping set the research direction in the initial stages and subsequently guiding me step by step to translate my research ideas into tangible outcomes, has been invaluable. Additionally, his support for my attempts to apply for doctoral programs during the thesis period has been immensely encouraging.

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1.Introduction

This chapter provides an overview of the study, including its research background, motivation, and the research gap it aims to fill. It also presents four guiding research questions for the investigation. Also, this chapter introduces the dataset and methods employed in the study. Finally, it outlines the structure of the thesis, providing readers with a clear understanding of the content of each chapter.

1.1 Background of the Study

Fan fiction is considered stories created by individuals who highly value the original work, crafting narratives based on existing storylines and characters. These works are seen as both subverting the original canon and injecting new vitality into classic texts (Thomas, 2007). Fans interpret the original work through the writing of fan fiction (Kaplan, 2006), thus not only explaining the character relationships observable in the original work but also elucidating the potential reasons for unofficial relationships (Thomas, 2011). Authors of fan fiction vary in style and genre, ranging from short stories to lengthy series, leading to different demands for character development. For instance, stories focusing on romantic relationships require different character portrayals than those centered on other types of relationships. Despite the diversity in these stories, they share the commonality of narrating events from the characters' perspectives. A sentence from the paper by Gray and Mittell (2007, p. 18) describes the motivation for the study in a good way, "a well-told tale lives and thrives after its telling". Therefore, this study aims to explore how character combinations and interactions are changed within fan stories.

Many scholars were curious about the extent to which fan fiction "borrows" or "deviates" from the related "source" text, engaging in discussions about the narrative relationship between the original canon and fan fiction. For example, Jenkins (2012) discussed several common rewriting strategies, such as expanding on aspects not fully explored in the original, correcting unsatisfactory plot points, and clarifying the purposes behind the revisions by analyzing specific fan fiction texts and the authors' intentions. Some researchers have also focused on character analysis. For instance, Brottagger et al. (2023) combined quantitative analysis with fan fiction theory to reveal trends in character transformations by examining name frequencies, co-occurrences, and sentiment analysis. Additionally, Thomas (2007) described different types of fan fiction found on various websites, showcasing the methods of adaptation.

These studies demonstrate that focusing on the transition from canon to fan fiction with different perspectives and analytical approaches can yield diverse insights. However, character transformations have always been a focal point of scholars' attention due to the importance of characters in narrative construction.

Morrow (1985) pointed out that readers construct their understanding of the narrative world by associating characters, events, and locations with the order in which this information appears in the narrative. Narratives involve not only the characters themselves but also their actions and interactions, as no character exists independently in a vacuum. Their existence and development are interdependent, collectively constructing the entire narrative system through interactions with other characters (Woloch, 2009, p.15). Characters and their relationships are the backbone of narratives (Cipresso & Riva, 2016),

and studying the sum of complex interaction dynamics between characters can help us better understand the dynamic integration of characters within narrative structures (Woloch, 2009, p.17), and interactions between the protagonist and surrounding characters (especially antagonists) drive the story's development (Lee & Jung, 2020).

Considering that authors invent and construct stories by portraying characters in various ways, discovering new relationships between characters (Card, 2010), we plan to continue focusing on characters in fan fiction. Due to the significance of character interactions in narrative research: characters drive plot development through their actions and influence narrative structure through their behaviors and positions (van Es & Schäfer, 2017). Therefore, this study primarily concentrates on character interactions in fan fiction.

In this research, we take each fan fiction as a story, with the observation of character interactions as the entry point. By contrasting these interactions with those in the original canon, we aim to use a network perspective to observe which specific characters or relationships fan authors choose to preserve, develop, or adapt from the original work, potentially indicating unique expectations regarding certain character relationships.

However, analyzing just one fan fiction may not reveal the diversity of how fan authors reshape or develop character relationships. Thus, we need to analyze multiple fan fiction works, necessitating the introduction of quantitative methods and intuitive visualizations to observe character relationships in stories. By incorporating network theory, we can not only address the aforementioned issues but also quantify plot elements to provide us with quantitative evidence.

As Moretti (2011, P.4) pointed out, network models allow us to discuss a character without being overshadowed by the richness of the character itself. The protagonist is simply "the character who minimizes the sum of distances to all other vertices". By using Network analysis, we can reveal the results of character connections within complex plots.

In a word, networks provide us with a visual way to extract the possibilities of character interactions from story structures and transform them into a set of images that can be easily understood on a plane.

Existing research has shown that applying a network analysis perspective in narrative analysis can utilize distant reading methods combined with character networks to simplify plots and address specific issues, such as detecting recurring patterns and narrative events, identifying character attributes (e.g., protagonists and antagonists), or the significance of characters (Labatut & Bost, 2019). For example, quantifying the co-occurrence of film characters in the same scene as relationships, analyzing role relationships based on network graphs, identifying key characters based on network features, and recognizing macro and micro character communities in the film (Weng et al., 2009). Furthermore, constructing character networks based on dialogue in films, identifying character importance by calculating network features, and recognizing main groups based on character connection weights and bidirectionality in the network (Park et al., 2012).

In addition to narrative analysis in films, similar methods have been applied to narrative research on textual stories, especially in understanding the characters.

Scholars have constructed a character network based on the types of social events in Alice in Wonderland and identified the functional roles of characters in the text, such as the

protagonist and the holder of the narrative perspective, by calculating network metrics (Agarwal et al., 2012). Other researchers have used computational methods to extract character dialogue networks from novels and evaluated literary theories related to character dialogue relationships of that period by calculating network features (Elson et al., 2010). Although similar networks were used, different analytical methods were applied. Carron & Kenna (2012) quantified the interactions, discussions, and co-occurrences between characters as relationships and used network analysis to compare the degrees of reality and fictionality in mythological narratives from three different cultures. Similarly, Pojoga et al. (2020) constructed character networks for novels, establishing connections between characters based on their dialogues, but they also explored character relationships and individual discourse structures by incorporating social class differences.

1.2 The research gap the thesis aims to fill.

Due to the vital importance of character interaction analysis in exploring storytelling, focusing on character interactions enables a deep observation of the similarities and differences in storytelling between the original canon and fan fiction. While existing literature has utilized network analysis to delve into story plots and character understanding, there is a tendency to evaluate the narrative constructions between the original canon and fan fiction.

Furthermore, these studies primarily focus on novels, plays, and films, rather than simultaneously examining character interactions in films and fan fiction. Lastly, in terms of methods for constructing character networks, existing literature relies on character co-occurrence or dialogue, presenting relatively limited approaches. Therefore, this study aims to further enrich research in the relevant field through the utilization of character networks in these three aspects.

Regarding the research objectives, this study not only aims to analyze character relationships in stories from a network perspective but also seeks to identify common character pairing patterns using this approach. Through this, it aims to explore how fan authors fill in the "gaps" in the original canon by reshaping characters in their created stories.

Regarding the research subject, the dataset under study comprises 97 fan fiction works, which were identified by searching and filtering on the AO3 platform based on tags related to the original canon that we aimed to focus on (specific filtering criteria detailed in the next chapter). Fan fiction, distinct from cinematic narratives and other text types, constitutes a unique literary form (Geraghty et al., 2022). Therefore, this study's construction of character networks in fanfiction adds diversity to the application of narrative analysis through character networks.

Regarding the research methodology, we have diversified the methods for constructing character networks. During the process of extracting data on character co-occurrence within the text, we expanded the definition of "co-occurrence." Apart from establishing links between characters in the network based on the co-occurrence of character names within a certain textual window and dialogues between characters, we also constructed a network of character interactions based on the verb headwords in sentences describing characters. Therefore, in handling text data, we not only employed NLP to identify and extract character entities from the text but also utilized Spacy to analyze the sentence structure in the text to extract data for constructing character networks. Finally, we

integrated this data using network analysis libraries in Python and visualized it in network form.

Moreover, in our network analysis process, we combined image clustering with computing network features. Considering the multitude of networks, we aimed to apply a "distant reading" approach to analyze the visual structure presented by the networks, identifying similarities and differences. Visual analysis plays a role in enriching perspectives in this study, so we primarily computed network features and analyzed the data results. In terms of data statistics, we calculated the number of nodes in the network, network density, and the centrality of nodes, including degree centrality. Also, we computed the frequency of connections between specific roles in the character network and analyzed the results to identify important characters in the story and common character pairings.

In addition to innovating in the three aspects mentioned above, we will also integrate conclusions from existing studies to discuss the findings of this research. Based on scholars such as Xu et al. (2011), Milli & Bamman (2016), and Thomas (2011), fan fiction often focuses more on secondary characters than on main characters. Therefore, our study will contrast and discuss the transformation of main and secondary characters from the original canon to fan fiction, in comparison with existing research findings.

1.3 Research Question:

To achieve the research objectives, we place the main research question of this study: What is the difference in terms of creating character networks between original movies and fanfiction?

Throughout the research process, we will gradually piece together the answer to the main research question by addressing the following sub-questions:

1. Which characters in the original film have the highest interaction frequency, and what are the most common character pairings?
2. Are there differences in character pairings between fanfiction and the original film?
If so, what are they?
3. What are the most common interaction patterns among these character pairings?

1.4 Thesis structure

Next, we will outline the structure of the article, detailing how each section contributes to the overall objectives.

The first chapter will introduce the research background of the study, helping readers understand the main content and objectives of the study. The chapter includes the research questions and objectives, as well as a description of the article's structure and how each

part contributes to addressing the research questions.

Chapter Two will be the literature review. This section is divided into two parts: the first part introduces the concepts and theories relevant to this study, including character, fan fiction, and network analysis. The second part discusses research related to these theories and concepts, such as studies on transforming original works into fan fiction and the application of network analysis in literary analysis.

In Chapter Three, we will detail the composition of the database and the methodologies used. This includes methods for data collection, character network construction, and network analysis.

In Chapter Four, we will interpret the results of network visualization and the calculations of network characteristics. In this part, we focus on discussing the information about characters in fan stories reflected by the character networks constructed using different methods.

In Chapter Five, we will summarize the research results by answering the main research questions and sub-questions. Subsequently, we will discuss our results in the context of relevant literature. Then we will reflect on the methods used for constructing character networks, along with the limitations and constraints present in this study. Finally, we will summarize potential directions for future research.

2.Literature Review

In the following content, we will divide the literature review into two main parts. The first part focuses on theories relevant to this study, including definitions of characters and research on character interactions in narrative studies, the definition of fan fiction and related theories, as well as an introduction to network analysis theory and what character networks entail. The second part will review literature closely related to concepts involved in this study, and in conjunction with the above-mentioned literatures, clarify the research gaps that this study aims to address. Finally, we will position this study within the field of digital humanities to underscore its significance and relevance.

2.1. Theories and Concepts Relevant to This Study

2.1.1 Characters

What is a character? Scholars define "character" from different perspectives based on various fields. Frow (2014) regards "characters" as both a rhetorical device and a form of image representation, emerging from a narrative context as figures that generally represent human shapes or forms. Jannidis (2013) defines characters as participants in story worlds created by various media, contrasting them with individuals in the real world. Characters are fictional people in novels created by authors, forming part of the fictional world (Card, 2010). Characters are distinguishable fictitious entities capable of thought and action (Frow, 2014). Research on the concept of characters has sparked various discussions, shaping multidimensional studies of characters. It is worth emphasizing that characters, to some extent, are fictional groups expressed through language. Studying the language of individual characters alone is limited, as it does not account for the interactions and functions among characters that also affect their representation (Smeets, 2021). To gain a deeper understanding of the role characters play in the story, we need to explore their function within the narrative.

2.1.1.1 The Role of Characters in Narrative

The story world is gradually constructed during the narrative communication process, and characters become one of the key elements that inspire and shape the narrative (Frow, 2014). Characters not only contribute to the shaping of the story's themes but also enhance readers' understanding, feelings, and evaluations of the characters through the depiction of their decisions and actions (Hogan, 2010). However, the role of characters in the narrative is multi-layered, involving both their significance (such as main characters, secondary characters, and incidental characters) and how they function within the narrative structure. Therefore, we still need to consider the relative importance or marginality of characters in the story (Garvey, 1978). Furthermore, the relationships between characters and their roles in the narrative are closely interconnected. Characters not only drive the development of the story but also influence readers' and authors' evaluations on ethical and emotional levels (Hogan, 2010).

In summary, characters in the narrative each serve different functions, with various levels of characters playing unique roles in the story. Above literatures provide a crucial analytical dimension for our upcoming character network study, focusing on how characters perform their different functions within the narrative.

2.1.1.2 The Role of Character Interactions in Narrative Storytelling

Several scholars have explored the impact of character behavior on the narrative, emphasizing that character actions are closely related to the events within the story. Aristotle (as cited in Heidbrink, 2011, p. 79) believed that the function of characters is primarily manifested through their actions, with actions being more important than the personalities of characters. Marvin Mudrick (as cited in Heidbrink, 2011, p. 74) further pointed out that character behavior and decisions not only propel the development of events but that events and the narrative also, in turn, influence the shaping of characters. Tzvetan Todorov proposed (as cited in Heidbrink, 2011, p. 83) that the priorities of characters varies in different types of narratives. Character traits were both the causes and the results of behavior, their traits influenced their actions, and these actions, in turn, further shaped and revealed their traits. The above perspectives help us understand the crucial role of character behavior in the narrative, illustrating the bidirectional influence between character actions and narrative events. This provides a theoretical foundation for understanding the reshaping of fan fiction through character network visualization. By analyzing the construction of character behavior, we can further reveal how stories are rewritten.

Moreover, Moretti (2011) argued that the plot is the arrangement of characters and their actions. The plot is also seen as the skeleton of the narrative, as it reflects the relationships between events and the connection of each event or element to the whole. This view further supports the research approach of understanding plot structures through visualizing character networks. Elizabeth Bowen's view (cited in Egan, 1978, p. 456), that "the plot is the story," further illustrates the unity between story and plot. This indicates that when we consider "What is the story of X?" we are essentially discussing "What is the plot of X?"—that is, the story is the actions understood by the reader. Since interactions between characters are a form of character behavior, network visualization of these interactions allows us to further explore the narrative of the story.

After exploring the importance of character interactions in understanding the plot, we turn our attention to fan fiction. Fan fiction, as a reuse of original narrative and characters, demonstrates how enriching and expanding the understanding of the original narrative can be achieved through creating a new story.

2.1.2 Fanfiction

2.1.2.1 What is Fanfiction

Fan fiction can be defined as stories created by those who usually have a deep appreciation for the source text, deriving from the settings, plots, and characters of an established narrative. Meanwhile, "fanon" refers to fan-created plots or settings, while "canon" denotes the official plotlines (Thomas, 2007) or the universe, background, and characters presented in the media source (Busse and Hellekson, 2014). However, definitions of fan fiction vary from different perspectives. Some scholars extend the definition of fan fiction to "archival" writing, emphasizing intertextuality and repeated engagement with the original text (Busse and Hellekson, 2014). Thomas (2011) defines fan fiction as stories created by fans based on plots and characters from a single source text or work's "canon," highlighting that fan fiction can be viewed as literary stories driven by fans. Some scholars define fan fiction as a part of the broader fan phenomenon, which has become an internet-based online phenomenon with the development of the internet in recent years (Uckelman, 2018). Jamison (2013) compares fan fiction to a large, complex bag within a world,

emphasizing the diversity of the community involved in creating fan fiction.

Piper (2015) views fan fiction as works created by media fans who claim their favorite text fragments and rewrite and re-present them in some way, thereby providing something for themselves and other fans who share their interest. These narratives created by fans often present existing story worlds in new and sometimes unconventional ways. Scholars argue that a common understanding behind the motivation to write fanfiction is that fans seek more original source material or wish to explore beyond canonical content. Therefore, when fans engage in creating their own stories, they may fill in missing scenes from the source text, extend event timelines beyond the canon itself, or replicate characters with whom they feel a connection. For those seeking more canonical content, this may involve pairing unrelated characters, focusing on secondary characters that are perceived as neglected in the canon, or completely altering the canonical context in a different universe (Pugh, 2005).

Jenkins (2012) coined the term "textual poachers" to describe and reassess fan activities. Fans satisfy their needs and desires by reusing the availability of popular culture, not just accepting the original works, but reinterpreting and playing with them, similar to musicians adapting original songs (Lee, 2011). Complex texts created by fans make full use of the diversity of existing fan works and canonical characters. These texts rely on a deep understanding of the source text and awareness of other fan works (Kaplan, 2006). The perspectives from the above literature help us understand how fans rewrite and reimagine segments from the original works to meet their needs, especially in how they fill gaps or expand storylines from the source text.

In this study, we primarily adopt Thomas's (2011) definition of fanfiction, considering fanfiction as stories created by fan communities based on the original canon. According to the perspectives of the scholars mentioned above, by analyzing the stories created by fan authors, we can understand the diversity of interpretations of the canon.

2.1.3 Network

2.1.3.1 What is Network

After clarifying the definition of fan fiction and its re-creation of the canon, we will next introduce the concept of networks and the elements that constitute a network. We will emphasize how network analysis serves as a visualization tool to analyze and interpret complex relationships. This will provide theoretical support for the subsequent application of network analysis to the study of fan fiction.

The etymology of the term "network" in Germanic and Romance languages involves assumptions about structure, patterns, order, and distribution. For scientists, a network is an abstract object that defines entities, or vertices, and the pairwise relationships between them (referred to as edges or links). In the humanities, networks are more commonly used as a metaphorical tool to provide a conceptual framework that helps us understand the fundamental ways of the world. This metaphor allows us to view one thing as another to reason and comprehend another conceptual domain. With the application in social sciences, social network analysis has developed a range of concepts and measurement methods that significantly shifted the metaphorical background of "network," viewing networks as something measurable mathematically (Ahnert et al., 2020, p.13).

In social network analysis, a network refers to a group of entities and their connecting

relationships. The elements constituting these entities can be people organizations, countries, or non-human entities like words. Relationships can take various forms, with different definitions based on different contexts, such as emotional (like friendship or trust), cognitive (like perceived friendships), flow (like trade or information), roles (like teacher-student relationships), and interactions (like email exchanges). Relationships can also be categorized as states and events, with state relationships being stable and enduring, while event relationships are transient (Scott & Carrington, 2011).

The literature indicates that the essence of a network lies in its emphasis on structure, pattern, and distribution. In this study, we will utilize these characteristics to examine the structure of character networks and analyze patterns and distributions of character combinations.

2.1.3.2 Network analysis

With the advent of the digital realm, metric methods in network analysis have been more widely applied, such as measuring network paths and centrality (Ahnert et al., 2020). In this section, we will discuss the applications of network analysis across various fields, illustrating how network analysis is used to understand "relationships" in different domains.

A network perspective emphasizes structural relationships, with the core objective of network analysis being to measure and represent these structural relationships and explain their causes and consequences. However, network analysis is not only a framework for describing how a set of entities are interconnected, it is also a theoretical approach to analyzing and interpreting a phenomenon. For example, Scott, (2012) pointed out that social networks refer to specific connections among specific individuals. Beyond these connections themselves, the collective characteristics of these connections can also be used to explain the social behaviors of these individuals. Thus, network analysis provides us with an observational perspective and a method to explain phenomena.

The mathematical foundation on which network analysis is based is graph theory, which includes axioms regarding relationships between a set of points and the lines connecting these points. Therefore, it focuses on patterns formed by points and lines and involves exploring these patterns mathematically or visually (Scott, 2012). In social network analysis, individuals are seen as "nodes," with their relationships seen as "edges" (Carrington & Scott, 2011). In relationship studies, scholars focus more on the roles of network members (e.g., friends), their emotional ties (such as likes and dislikes), cognitive awareness (such as knowledge of each other), and interactions. Generally, interactions are seen as behavior-based connections, and the ways interactions are measured (like the frequency of communication between individuals) and the ways emotional relationships are measured (like the degree of liking or disliking) are sometimes interchangeable (Marin & Wellman, 2011). In their research, network analysis scholars combine graph theory theorems to describe and explore social network characteristics. For instance, they distinguish between directed and undirected graphs, considering the direction of connections to represent asymmetric relationships or the flow of influence or resources between participants. Weighted graphs can assign values or "weights" to connections to account for the strength of relationships. "Nodes" can also be linked by multiple edges to represent different types of relationships (Carrington & Scott, 2011).

Based on the theories mentioned above, we can further apply network characteristics (such as centrality, density, and weight) to quantitatively describe the interaction patterns

between characters in fanfiction. This method helps us reveal the composition and function of characters and their relationships in narratives. Besides, quantitative methods facilitate comparisons between networks.

However, the interpretation of distance measurements depends on the analyst's judgment and research objectives. For example, in a friendship network, closer distances (like 2 steps) typically represent closer relationships, hence being more important than farther distances (like 10 steps), while in a credit network, the situation may differ as credit relationships may not heavily rely on physical or social distances (Scott, 2012). Therefore, higher weights do not necessarily indicate closer relationships. Depending on the relationships mapped by the network, the interpretation of network characteristics' calculation results varies, necessitating a critical approach when analyzing data.

Here are some common interpretations of network characteristics. First, analyzing vertex features in a network. For example, centrality measures are often used as indicators of power, influence, popularity, or prestige (Carrington & Scott, 2011). Betweenness centrality measures the frequency with which a single node lies on the shortest paths between other nodes in a connected network. Individuals with high betweenness centrality are more likely to act as bridges in the network. A bridging role in a network implies that the node plays a crucial role in the flow of information, controlling or influencing the flow of information from one part to another. These two structure-related features (centrality and betweenness centrality) are commonly used to identify 'opinion leaders' because this role not only requires specific attributes but also the correct network position that enables them to effectively disseminate information and exert personal influence (Liu et al., 2017, p.3).

From the perspective of overall network properties, network density allows us to quickly understand the tightness of a network (Ladd et al., 2017). The density of a graph is a direct measure of its cohesion, reflecting the level of interconnectedness between nodes in the network. If a network has a high density, it means that most nodes are directly connected, resulting in a tighter network of relationships. However, when used as a reference for comparing networks of different types, there are limitations, as density varies with changes in network size. Therefore, as the network size increases, although the theoretically possible number of connections increases, the actual number of connections may not increase proportionally, leading to a decrease in density (Scott, 2012, p.44).

Another strategy in network analysis is to focus on the differentiation within the network to reveal its structural diversity. This approach is commonly used when the connecting points in the network do not form a single connected whole but instead comprise multiple structurally distinct parts. In other words, a graph may contain one or more components of varying sizes, where a component is a part of the network in which all nodes are interconnected. These nodes within a component have no direct connections to nodes in other parts of the network. The number and size distribution of components are fundamental indicators for measuring network diversity as well as the flow of information and resources (Scott, 2012, p. 46).

Such analytical strategies enable us to observe the diversity of network structures and alert us to the fact that connecting points in a network may not always form a single cohesive whole. The literature indicates that network analysis can not only describe relationships between entities but can also use network features to explain these relationships. By applying the aforementioned methods, through analyzing centrality, we can identify which characters hold more influence or power in the story, and by analyzing edge weights,

we can understand the strength of relationships between characters. Network density can serve as a metric for comparison, used to quantify and compare the closeness of relationships between characters in different narrative networks.

After discussing the basic concepts of network analysis and how entities and their relationships within networks can be quantitatively analyzed using network features such as centrality, density, and edge weights, we will now focus on the "character network." The character network is a specific application within network analysis, particularly widely used in literary and narrative analysis.

2.1.3.3 What is a Character Network

According to Labatut and Bost(2019, p1), "A character network is a graph extracted from a narrative in which vertices represent characters and edges correspond to interactions between them". Similarly, as mentioned by Moretti (2011), a character network refers to a network where characters serve as vertices and interactions as edges, with characters and actions being integral components of the plot. The ways in which two characters are linked together can vary, such as through dialogues between characters or characters with lines in a scene. Both methods are reasonable but come with their own limitations, the identification of which requires discussion in the context of the analyzed work. Scholars also point out that constructing character networks can provide an intuitive way to analyze novel plots and characters, combining visual representations with story content to facilitate a more detailed study of relationships between characters (John et al., 2019).

In the first section, we defined the concept of characters, analyzed their roles in the story, explained the definition of fanfiction, and introduced network analysis theory. These theories provide the necessary background knowledge for understanding character interactions and network structures in fanfiction. In the second section, we will first explore the application of network analysis in literary studies, then review existing research on how fanfiction adapts original works, and finally discuss the positioning of this study in the field of digital humanities.

2.2 Studies related to main concepts.

2.2.1 Applying Network Analysis in the Literary Field

An increasing number of scholars are utilizing network visualization, social network analysis theories, and quantitative methods from network science to address their research questions. Humanists abstract their research subjects into data points that can be input into databases or spreadsheets, utilizing network analysis as a tool to describe given archives or corpora, making it a part of answering research questions. This approach is widely applied in the field of literary studies (Ahnert et al., 2020).

Scholar gather data related to texts, identify relationships and network members, and conduct analysis based on network structures. The character network of a given narrative (novels, plays, movies) simulates the structure formed by relationships within its character system (Woloch, 2009). For instance, Holanda et al. (2019) attempted to distinguish different literary genres by analyzing social network features in literary texts, proving that using character networks is not an ideal method for genre differentiation. However, through these features, they could identify communities of major and minor characters in the text and the frequency of connections between these communities.

Besides, John et al. (2019) visualized networks of characters and their relationships in a specific text, combining visual representations with novel texts to understand commonalities and differences in character constellations. Sims and Bamman (2020) modeled patterns of information dissemination in over 5000 British novels, identifying through network visualization which characters facilitate information flow, revealing that female characters often play more bridging roles in information dissemination than male characters. Segel and Heer (2010) introduced the concept of using network visualization as a form of digital storytelling, noting that such visual narratives can provide a quick and intuitive exploration of large amounts of data resources.

Ryan (2005) explored the idea of using network analysis and visualization software packages like Gephi as "creative systems," where the relationships between two characters symbolize their coexistence in the narrative portion. The entire set of relationships among all characters forms a formal model of the character system, aiding in its display and analysis. For instance, Moretti (2011) used network modeling to compare the importance of protagonists in Shakespeare's "Hamlet," while Waumans et al. (2015) defined novel story features based on social networks of novel characters.

In the network they constructed, nodes are defined as characters, and links are defined as dialogues between characters. They believed that this social network reflects distinctive features of each story, such as the number of main characters and the author's writing style (Xanthos, 2016; Perri et al., 2022).

In addition to using character networks as analytical tools, some scholars have shifted focus to the narrative capability of networks. Venturini et al. (2017) proposed a network narrative perspective based on visual attributes on the character network of the "Iliad."

Some scholars aim to utilize network models and social network analysis methods, applying quantitative approaches to understand narrative features. Gleiser (2007) constructed a character network based on Marvel Comics, where two characters being present in the same publication established a connection.

By computing network features, they sought to determine whether the fictional networks created by writers resembled real social networks to some extent or appeared more like random networks, demonstrating similarities with differences between the two. Alson Li et al. (2019) utilized complex network theory to analyze character social networks in literary works, offering a perspective for literary analysis. They showed that the character social networks in "Harry Potter" and "The Lord of the Rings" exhibit small-world and scale-free properties, suggesting that a possible explanation for the popularity of these two literary works lies in the fact that the character social networks in the stories follow the principles of complex network theory observed in real-world social networks, making them more relatable to audiences.

Smeets (2021), combining network analysis tools with 170 contemporary Dutch-language novels, explained the centrality of characters, communities, and conflicts in fictional social networks. This integration of social network analysis with narratology showcased how descriptions and representations of social themes in Dutch novels are presented. This work offers a new perspective, suggesting that through social network analysis, we can delve into the centrality and community relationships among characters in texts.

However, in addition to discussing narratives through network analysis, some scholars have specifically explored the impact of different methods of constructing character

networks on research. Elson et al. (2010) built social networks based on character interactions in 19th-century British serialized novels, primarily linking characters based on their interactions in dialogues.

They examined the correlation coefficients between network features (such as the number of character vertices and the average degree per vertex), showing that the shape of social networks in novels is more influenced by the narrative style than by the novel's setting or the number of characters.

Coupette et al. (2024) created a dataset that offered various representations of relationship data extracted from Shakespearean plays, presenting the data using different network graphs ranging from simple co-occurrence graphs to hypergraphs that incorporate complex communication settings and character contributions. The emphasis of different network visualizations varied, demonstrating how different data representations can provide unique information while also obscuring certain details. Similar to this work, John et al. (2019) introduced various visualization methods for analyzing text documents of different scopes, including using network analysis to visualize character relationships in terms of quality and transitions. Analysts could switch between two different graphical representations, adjacency matrices and node-link diagrams, to display and understand the overall situation of characters and their relationships. Bonato et al. (2016) constructed character networks for three novels, "Twilight," "The Road," and "Harry Potter and the Goblet of Fire," and analyzed them along with existing character networks from corresponding movies. They compared actual character network data with simulated network data generated from various random complex network models and found that the Chung-Lu model was the most suitable for describing character networks.

Although the research objectives of the aforementioned studies differ, their experimental results indicate that character networks can to a certain extent map the simulated character relationships in the given narratives.

2.2.2 Research on the Comparison between Fan Fiction and Original Canon in the Field of Fan Fiction

When studying fan fiction, ethnographic methods are often used to gain a deeper understanding of the background and culture of the research subjects, considering fans' motivations and combining interpretations based on comments about fan fiction itself (Van Steenhuyse, 2011). For example, Fathallah (2017) aimed to address how fan fiction alters our experience of media content and generates different categories of social interpretation. This study not only focused on how highly praised fan works adapt the original texts but also considered the influence of the original authors and various fan communities on the formation of discourse.

In addition to integrating fan fiction with cultural studies, many scholars have conducted literary analyses of specific fan fiction stories. But most existing work is qualitative and descriptive rather than quantitative. Common methods include reading fan fiction and analyzing fan comments related to it (Barnes, 2015). According to some research result, fan fiction often expands storylines by focusing on underexplored themes or minor characters from the original story, or by improvising on the content of the original work (Bond & Michelson, 2008; Jenkins, 2012). Many studies have analyzed Harry Potter fan fiction, revealing imaginative backgrounds for Harry's student nemesis, Draco Malfoy (Stein & Busse, 2009; Lanier & Schau, 2007). These fan fictions adopt Draco as the narrative perspective, explaining character motivations through the actions he takes.

Furthermore, under the guidance of fan fiction authors, Malfoy transitions from a villain in the original text to a hero in his own story (Lanier & Schau, 2007).

Moreover, scholars have observed that fan fiction often responds to significant plot developments in the original story by rewriting narratives to either support or challenge these plot points. For instance, some fan fictions imagine a character never died or that two characters never separated (Stein & Busse, 2009). Some scholars argue that fan fiction serves as a way to adjust the relationship between readers and fictional characters, leveraging readers' familiarity with the original characters to evoke emotional responses, which explains why many fan fictions offer a high level of immersion (Van Steenhuyse, 2011; Berkowitz, 2013).

Fan fiction also often focuses on relationships between characters. Fans who identify as "shippers" (derived from the word "relationship") support their favorite couples, who may not have been depicted as romantic partners in the original text (Barnes, 2015). Some scholars point out that romance in fan fiction differs from mainstream romantic stories. For example, Tosenberger (2008) noted that romantic fan fiction does not necessarily require a happy ending typical of traditional romance novels but rather strings together moments of tenderness in the story, focusing on conveying an emotional experience—either heartwarming or anxious—that may lack a concrete plot. Similar viewpoints are also mentioned by Van Steenhuyse (2011), who argued that fan fiction adapts the form of traditional romantic stories. By analyzing the portrayal of romantic relationships in House fan fiction, the author found that the fan works not only depicted the desires between character relationships but also explored how the protagonists overcome obstacles to build their romantic connections, without concluding with marriage.

Another popular type of fan fiction is center on one character experiencing severe physical or psychological trauma and then finding healing through the assistance of another character, strengthening their relationship (Stein & Busse, 2009). Given the concentrated focus on emotions in this type of storytelling, such narratives often deepen the emotional impact of the original work (Jenkins, 2012).

The adaptations mentioned above center around characters and their relationships, as well as the characters' social interactions, highlighting the prominence of characters' relationships in fan fiction.

Lanier and Schau (2007, p.330) analyzed three popular Harry Potter fan fictions, examining them in terms of "action (what is done), scene (where it is done), agent (who does it), agency (how it is done), and purpose (why it is done)". They found that compared to the original Harry Potter novels, the analyzed fan fictions paid more attention to the relationships between agents or the motives behind their actions rather than the actions of individual agents.

Jones (2002, p.86) suggested that "for fans, what matters most is not how characters move the plot forward, but what the narrative events reveal about the characters". Interestingly, some scholars propose that the central focus of fan fiction lies not only in characters and their relationships but particularly in their inner lives and emotions (Barnes, 2015). In alignment with this, a fan fiction author interviewed by Lee (2011, p.257) defines fan fiction as a way of "entering into the minds of beloved characters".

In other words, while fan fiction is often described as "filling in" gaps in the original story, exploring possibilities that were never seen in the canon, the specific gaps addressed and

potentials explored often involve characters' backgrounds, motivations, emotions, and desires. This imaginative engagement with fictional works seems to involve and may require two key factors of interest: a strong emotional investment in the source material, especially its characters and relationships, and a resistance or willingness to subvert the original text (Lee, 2011).

Therefore, fan fiction authors do not merely use the source material as a template for stories. Instead, their narratives focus on the characters from the source material, may not follow key plot developments from the original work (Stein & Busse, 2009). And may challenge the moral or value frameworks set in the canon by reshaping character actions or storylines (Jenkins, 2012), shifting the focus of different story elements from the original work (Lanier & Schau, 2007), rejecting flat portrayals of supporting characters in favor of providing complex or nuanced characterizations (Jenkins, 2012), and showcasing relationships that the original work either could not or would not acknowledge (Tosenberger, 2008; Larsen, 2011). Therefore, viewing writing fan fiction not only as a way to understand beloved characters but also as a form of advocacy for these characters (Jenkins, 2012).

The above studies suggest that fan fiction may reflect audiences' imaginative engagement with fictional characters, thus tending to emphasize or adapt from the perspectives of character relationships and psychological states. Authors may use fan fiction to fill in the missing parts of character backgrounds, relationships, motivations, and emotions left unexplained in the original canon. They may resist the original author's intentions, especially when these intentions contradict their own understanding or ideas about the characters (Barnes, 2015). The established results provide a theoretical foundation for our subsequent understanding of the strategies in fan story creation by combining network analysis results.

2.2.3 Quantitative Study of Characters in Fan Fiction

Scholars have explored characters and their relationships in fan fiction quantitatively. (Rowe et al., 2021) utilized Word2Vec, an algorithm developed by Google to discover which words are closest to target words, for text mining on 450 fan fiction works based on J.K. Rowling's "Harry Potter" series. The aim was to reveal certain characteristics or attributes of a specific character by examining the relationships between character names and other characters. The study found that fan fiction writers emphasize Hermione's femininity more than Rowling does, even though Rowling's wording allows for different interpretations. This research was able to uncover that fans share specific interpretations of Hermione in their fan works.

Milli and Bamman (2016) applied computational methods to fan fiction to quantitatively address differences between fan fictions and their source material. They used Booknlp to extract and compare characters from 10 canonical works and their corresponding one hundred stories, confirming that fan fiction authors tend to provide more in-depth descriptions of secondary characters from the source material and focus more on female characters. Also, Brottrager (2023) conducted a quantitative analysis of fan fiction mechanisms, features, and metaphors to illustrate trends and tendencies in character transformations from canonical works to fan fiction. By comparing the frequency distribution of character names, the result demonstrated that fan fiction indeed shows a distinct focus on specific characters, differentiating from the original work.

In their research, they demonstrated the focus on specific characters in fan fiction by

comparing the frequency distribution of character names, analyze character co-occurrence to highlight differences in the description of their relationships, and identify which pairings have gained or lost the attention of fan fiction writers. They also use sentiment analysis methods based on word embeddings to quantify emotional changes in characters, identifying certain types of characters destined to be re-conceptualized, such as minor characters in prequels or sequels, and re-designed villains following the typical fan fiction trope, such as 'enemies to lovers'. The study demonstrates the potential of using various computational methods to analyze fan fiction and highlights how the application of network analysis can reveal character relationships within the stories.

Kleindienst et al. (2020) also employed similar computational methods to explore how the fan fiction community uses source material and deviates from it in stories. They created a corpus of over 7,000 fan fictions from the TV show "Supernatural" scripts and the "Archive of Our Own" platform, focusing on changes in language, characters, themes, and emotions in the stories. By comparing the frequency of appearances of major TV show characters, they found that male labels dominated the corpus, and quantitatively observed the prevalence of homosexual content in fan fiction, echoing the findings of Tosenberger (2008). Schmidt et al. (2021) shifted perspectives, utilizing computational methods for social network analysis of crossover fan fiction to explore how fans alter and combine original material. They conducted social network analysis using the Gephi tool, discovering that original characters are a significant part of crossover fan fiction, with characters in fan circles largely clustered based on genre or national backgrounds.

2.3. Gap in Literature Review

In the content above, we reviewed the quantitative research on fan fiction, gaining insight into the existing quantitative methods and their applications. However, due to the limitations of the applied technical methods, such as the use of a single digital analysis tool, the research results only offer one perspective on how fan fiction is rewritten. For instance, merely relying on word frequency analysis fails to fully reveal the network of relationships between characters and focusing only on specific characters and their specific attributes in quantitative research neglects attention to the roles and functional positions of characters within the narrative. While some scholars have focused on ways character relationships change, the corresponding methods fail to observe specific character relationships within the overall narrative, thus uncovering implicit relationships between characters (such as bridge characters, key nodes in social networks).

In order to enrich the analytical perspectives brought by quantitative research methods, this study will employ network analysis as the primary analytical tool and will utilize three different methods to construct character networks. Network analysis can not only reveal the complex relationships between characters as a whole but also illuminate how these character relationships construct narratives. This is because network analysis focuses not only on the attributes of individual nodes (characters) but also on the patterns of connections between these nodes and how they form communities.

This study aims to provide a new tool and perspective for fan fiction research by utilizing network analysis. Network analysis will not only help us understand the character networks in fan fiction but will also offer another perspective on exploring character interactions and relationships.

2.4. Positioning of This Study in the Digital Humanities

Due to the scale of digital data and the primarily investigative nature of its applications, the application and commentary on network research mainly occur in related fields such as computer science. However, currently, humanists are utilizing similar network visualization and quantitative analyses to elucidate cultural studies (Ahnert et al., 2020). In recent years, Digital Humanities (DH) has become a research area widely discussed among computational linguists, humanists, and visualization researchers. A typical task for humanists in the field of literary studies is character analysis in novels. For instance, they may aim to depict the main characters in a plot, gather information on character relationships, or compare different character constellations and plot events (Chen et al., 2023). Luczak-Roesch et al. (2018) introduced a novel tool for digital humanities research that utilizes network science and visual analysis. Their findings indicated that this approach fostered a new collaborative methodological practice, a blend of close and distant reading. Furthermore, information technology can aid digital humanists in uncovering complex social networks of characters hidden within large historical texts. Still, it is essential to recognize that a network is an abstract object, a collection of pairwise relationships (referred to as "edges," "links," or "arcs") between defined entities (referred to as "nodes" or "vertices"), prompting an awareness of the abstraction process: what information is prioritized and what information is abstracted away (Ahnert et al., 2020).

Therefore, we consider network analysis both as an opportunity and a challenge. The opportunity lies in network visualization providing an additional means to generate, explore, and analyze information compared to narrative argumentation or mathematical equations, valuable in both humanities and sciences. The challenge arises when deciding how to encode information about the relationships between entities (the "edges") and what level of information to include in mathematical measurements of the network, with more information being lost as we progress. In the humanities, critics of quantitative methods often focus on this "information loss," a critique less common in the field of network science starting from abstract networks (Ahnert et al., 2020).

From this perspective, this study not only serves as an experiment in using network science and visual analysis for textual content analysis but also as a practice of a mixed method of close and distant reading. It will also focus on the process of abstracting related character information in fan stories to generate character networks, examining what information is gained and what is lost in this process, enriching reflections on quantitative methods in relevant fields.

3.Data and Methodology

This chapter elucidates the research data utilized in this study, the construction of character networks, and the methodology for analyzing these networks. Firstly, it outlines the process of data collection, specifying what data was gathered and why. Next, it provides detailed information about the methods of data processing, specifically how character networks were constructed. Finally, it introduces the strategies for analyzing the networks and the corresponding methods.

3.1. Fan Fiction Data Collection

3.1.1 Canon Selection:

When discussing fan fiction, the term 'canon' becomes essential. Understanding the canon, i.e., the events presented in the source material, which provide the universe, background, and characters, is crucial for dealing with fan texts. Additionally, fan fiction, created by specific fan communities and recurring throughout the fan text, are pivotal. For instance, for fans of "Star Trek," typically the four TV series and any movies are considered canon, excluding animated TV shows or novels. Understanding the canon is especially important for creators of fan texts as they are judged based on how closely they adhere to or diverge from it (Kristina & Karen, 2006). Therefore, clarifying the canon of the fan fiction texts under study is necessary. This not only helps establish clear research objectives but also provides standards for data collection. In this work, we have chosen the movie version of "The Rocky Horror Picture Show" as our canon for the following reasons:

First, in terms of its content, it is an intertextual film, with its non-traditional setting and multi-layered interpretations offering fans extensive creative freedom. "The Rocky Horror Picture Show" is a cult film that tells the story of a young couple who stumbles upon the home of Dr. Frank-N-Furter. The castle's owner, Dr. Frank, is a provocative transvestite who invites Brad and Janet to his party, leading to a series of absurd and bizarre events within the castle where Frank resides. Unlike mainstream films, it involves atypical situations of typical characters, sympathy for deviance, challenges to traditional authority, and reflections of social pressures. Furthermore, the film subverts traditional appearances and sexual meanings of characters, challenging conventional standards. It is this non-conformity to traditional norms that allows the film to be interpreted in diverse and even contradictory ways. Hence, being a film that blends various genres and exhibits high intertextuality, its multi-layered complexity makes it an ideal subject for fan fiction exploration and reinterpretation (Grant, 1991).

Second, alongside this, being a fan of this film, my familiarity with its plot is one of the main reasons for choosing fan fiction based on this movie as the subject of study. This familiarity with the film's plot and characters can aid in accurately identifying the characters and relationships preserved or modified in fan stories.

Third, considering of dataset size, on the ao3 platform, the number of fan stories corresponding to this movie is relatively low, with only 776 related search results. Compared to fan fiction categories with massive numbers, this category is not mainstream. Studying fan fiction within this category can help balance research perspectives, increase diversity, and offer supplements to existing popular categories.

3.1.2 Data Acquisition Platform Selection

Considering the convenience and accessibility of data collection, we have chosen to gather

the fan fiction data we need on the Ao3 platform, a fan fiction website. The Ao3 platform provides rich and easily accessible data resources (Pianzola, 2020), allowing us to obtain the necessary raw data. This platform is a fan fiction archive that uses open-source archiving software. The data on the Ao3 platform is freely accessible, and we can use the data from the Golem project database, ensuring both the ease of data accessibility and the transparency and openness of the data. Furthermore, authors uploading stories on this platform can add various tags. Through these tags, we can filter out fan fiction related to the original source material we want to study. The character tags and relationship tags on this platform list the possible name variants of characters that may appear in the stories, as well as the relationships in which story characters are involved. This makes it easy for us to filter fan fictions related to the source material we are studying. This detailed tagging system can help us select different fan fiction texts based on character relationships.

3.1.2 Story Selection Criteria and introduction of Corpus

Since we focus on stories with specific tags and the study primarily revolves around characters, we selected stories based on character relationships to include the main characters more comprehensively from the movie and their various possible pairings. To achieve this, we first identified fan stories tagged as "The Rocky Horror Picture Show" using the search bar. Subsequently, we utilized the story character relationship filtering feature on the Ao3 platform. We randomly selected 20 stories from different character relationship pair tags. According to Ao3's explanations of relationship tags, stories under a particular tag involve the character relationships indicated by that tag. We downloaded stories tagged in the format "---/---," representing a romantic relationship between two characters, selecting 20 stories from each tag. We then filtered out stories that appeared in different tags (meaning a single story may contain multiple character relationship combinations and thus may appear under one or more tags), ultimately selecting 97 stories for research purposes. The tags we used are listed below: Magenta/Riff Raff (Rocky Horror); Brad Majors/Janet Weiss; Columbia/Eddie (Rocky Horror); Frank N. Furter/Everyone; Frank N. Furter/Brad Majors; Frank N. Furter/Janet Weiss; Columbia/Magenta (Rocky Horror).

3.1.3 Building Character Name Dictionary

To identify characters more accurately in fan fiction stories, we constructed a character name dictionary. This dictionary enables us to quickly retrieve names in subsequent analyses and recognize different references to the same character in a story as the same individual. Initially, we generated a list of character names based on the names appearing in the character relationship tags selected from the Ao3 platform, combined with names from the movie character network. However, upon inspecting the story content, we found that different names were used for the same characters across different stories. To address this, we utilized similar tags under individual character tags on the Ao3 platform to construct a character name dictionary containing variations of character names. The dictionary includes 11 English names and their variants, stored in a JSON file format.

3.1.4 Building Verb Types Dictionary

To map verb types to weights accurately, we constructed a verb type dictionary and assigned scores to these verbs. The idea behind scoring verbs is based on the author's mention in the paper of a character's direct actions towards another character, such as thinking about someone or actions performed jointly by two characters (e.g., fighting).

Additionally, distinct from descriptions of dialogue interactions in the story, action descriptions exist outside of dialogue narratives (Labatut & Bost, 2019). Therefore, we envisioned using action types as a way to detect interaction types, allowing us to more comprehensively identify character interactions described in the text.

To achieve this goal, we not only need to identify the actions taking place but also the characters experiencing these actions. To simplify the extraction process, we utilized the Spacy model to analyze the structure of these sentences and extract all verbs marked as central words to build a verb dictionary. The reason for selecting verbs as central words is that they represent the core of sentence structure and there is a systematic relationship between the root form of words and event structures. In other words, the root form determines the structure and features of the action or event represented by the verb in language (Levin, 2005). Therefore, we believe that by identifying the central verbs in sentences, we can reflect the ways characters interact.

After processing the 97 stories in the database using spaCy, we obtained a total of 2763 verb results. However, considering that some verbs only appear in a single text, we decided to simplify the data by removing verbs that appeared only once. In the end, a total of 1318 English verbs were included in the classification process.

When categorizing verbs, we took into consideration prior relevant research where few authors calculated scores to represent interaction (Labatut, & Bost, 2019). Agarwal et al. (2012) distinguished between two types of behaviors: one where the object character is unaware of the action taking place, termed as unilateral actions (e.g., missing someone), which they referred to as observational behaviors, and the other being all other behaviors. Trovati and Brady (2014) manually identified three categories of verbs and assigned signed scores to other verbs based on the categories: friendly, hostile, or unknown, without further detailed subdivision for the third category of verbs. Srivastava et al. (2016) utilized sentiment analysis to leverage the semantic of the textual context and differentiated interactions as cooperative or adversarial.

Combining existing verb classifications, when categorizing and scoring verbs, we considered not only the role of verbs in conveying emotional expression but also in describing actions between characters. Drawing from the extension of verb classification by Kipper (2008), where verbs describing interactions between two living entities were categorized, such as verbs like "encourage," "fight," "debate," we established our classification approach. The specific classification method is described below.

After reviewing the verb corpus, the most frequently occurring verbs were "smile," "look," "said," and "ask". Based on their frequencies and by leveraging the semantics of verbs, we categorized verbs and defined different types of interactions. According to the actions described by the verbs, we categorized them into five types: dialogue, affection, mental activities, facial expressions, and actions. During the scoring process, we planned to assign scores based on the directness of interaction conveyed by the respective verbs.

Verbs indicating affection were assigned the highest score because they convey direct and positive interactions between characters. Next, we assigned the lowest score to verbs related to mental activities because when engaging in mental activities, we consider the executor and recipient of the action to be the same individual, thus indicating lower directness of interaction (Korhonen & Briscoe, 2004). Specifically, verbs expressing concrete and direct interactions received the highest scores, while verbs involving mental activities and abstract expressions like facial expressions received the lowest scores.

Here is a detailed description of how verbs were categorized and scored: Words expressing strong positive emotions and intimate interactions were given the highest score (5), such as "love," "care," "like," categorized as affection. Next were words categorized as actions, marked with a score of 4. Verbs in this category depict specific actions and behaviors in daily life, including interactions, daily activities, and other emotional expressions, such as "move," "find". Words related to language and communication were grouped under conversation, with verbs in this category scored as 3, emphasizing the expression of opinions and emotions through speech, such as "yell," "argue," "roar." We also categorized verbs indicating facial expressions under facial expression, scored as 2. Lastly, verbs representing mental activities, thoughts, and abstract concepts were grouped as mental action, with verbs like "think," "wonder," "miss" scored as 1. The scored verb dictionary was saved in a JSON file format.

3.2. Collection of Movie Character Network Data

The network we selected for comparison is the character network of "The Rocky Horror Picture Show" in the "Movie dynamic" database. This dataset is one of the research outcomes of the article "Understanding Gender Disparities in the Film Industry Using Data Science". It is the largest open movie social network dataset, containing 15,540 movie dynamic networks. In this dataset, the movie network file of "The Rocky Horror Picture Show" includes the following data: character names and the links between characters. These links are established based on the time intervals between the appearances of two characters in the movie. If two characters appear within a certain time interval, a link between the two characters is created, with the weight of the link calculated based on the co-occurrence of the characters. In this article, they assume that two characters appearing successively in a short period may be related and simulate the interactions between these two characters based on this assumption (Kagan et al., 2020).

3.3 Methods for Constructing Character Networks

3.3.1 Method One: Co-occurrence of Character Names in Text Windows:

This method is inspired by the construction method of movie character networks, where the simple co-occurrence of two characters in a scene is often sufficient to infer their interaction. Simultaneously, we aim to employ the most straightforward method for detecting co-occurrences during the initial phase, which involves breaking down works into smaller narrative units and assuming that characters appearing together within the same unit interact (Labatut & Bost, 2019).

Various scholars have employed different approaches to dividing narrative units based on different forms of text. This can include structural divisions (such as by chapters) or length-based divisions (according to the number of sentences, paragraphs, or pages) (Labatut & Bost, 2019). Considering that different division methods can impact the detection of narratives differently, and acknowledging that fan stories vary in text length, we segment a story's text into text windows. When character names appear together within a window, a link is created between them, and each such occurrence increments the link weight between the two characters by one.

In our implementation, we set the length of each text window to be a quarter of the corresponding article's text length to accommodate varying story lengths. This adjustment ensures flexibility in adapting window sizes based on different text lengths. Furthermore,

drawing inspiration from the construction of movie networks in the applied database, we assume that characters appearing successively within a certain text window are related. For instance, in story #37372630, Riff Raff and Frank appear in two consecutive sentences within a paragraph, indicating a possible interaction between them within a certain text window.

During the network construction process, situations arose where networks could not be built due to the following reasons: First, some stories mix characters from outside the movie, making it impossible to identify all characters solely using a name dictionary. Second, some stories are very short, consisting of only three or five sentences and do not feature characters from the dictionary.

3.3.2 Method Two: Constructing Character Networks Based on Dialogue

We construct the network based on whether dialogue occurs between characters. Dialogue between characters is a crucial component that drives the development of the story plot, and verbal interactions can be assumed to signify co-occurrence, as two characters need to be present together to converse (Labatut & Bost, 2019). During the practical implementation, we found that the results obtained through code analysis did not align with our expectations. One of the reasons for this discrepancy is the diversity in how fan fiction authors refer to characters. Some authors do not directly use character names or pronouns but describe characters based on their attire in the movie. Hence, we believe that manual annotation can more accurately identify character dialogues.

We have established guidelines for the manual annotation of character dialogues. A link is created between characters for each instance of dialogue between them, with the weight of the link representing the number of dialogues. Here, we define dialogue count as the total number of sentences spoken throughout the entire conversation. For example, if Character A speaks 5 sentences to Character B in Scene C, and in response, Character B speaks 4 sentences to Character A, the weight between Characters A and B would be 9.

The specific reasons for not being able to construct the network are as follows: First, there is no dialogue in the story. Second, there is no actual text, only author descriptions and third, characters appear who do not exist in the dictionary.

3.3.3 Method Three: Constructing Character Networks Based on Verbs

In this method, we refer to the detection of interactions based on actions as mentioned in (Labatut & Bost, 2019), which considers the direct actions one character performs towards another. Some scholars divide the connections between characters based on the sender and receiver of actions. In this study, to simplify the extraction process and facilitate comparison with the two aforementioned methods, we consider setting undirected edges between characters in the extraction of graph data when there is interaction based on actions.

Initially, we map the character variants from the dictionary to the text, filtering out all sentences containing two or more character names. We apply a Root Verb dictionary to these extracted sentences. If verbs from the dictionary are identified in these sentences, a link is added between the characters mentioned in that sentence. Subsequently, we apply a verb score dictionary to the identified sentences, with the sentence score serving as the weight of the link. This method allows for a more intuitive analysis of the interaction strength between characters. For instance, a higher interaction weight between two

characters suggests that their link is composed of verbs with high scores, indicating a potentially close relationship between them.

Cases where character networks cannot be constructed include: First, sentences in the text primarily describing characters using pronouns. Second Usage of character names not found in the dictionary. Third, Narration from a first-person perspective with short articles lacking descriptive action narratives. Fourth, Stories consisting solely of dialogue without descriptive depictions of character actions.

Since language in narratives indeed constructs storytelling by encoding various real-world events or actions, including perceptions, thoughts, etc., and applies different types of verbs to different participants in the roles, such as agents, victims, or observers (Herman, 2007). Therefore, considering the construction of character networks based on the classification of different verbs, we envisage that this method may reveal different types of character relationships. For instance, we might observe character networks based on action verbs in some stories, while in others, similar networks cannot be constructed. By analyzing the weights of the links in the network, we can identify differences in character interactions. Some character combinations may exhibit higher weights in networks based on affectionate categories, while showing lower weights in networks based on action or mental activity categories, or even no links at all.

Cases where character networks cannot be constructed include: First, most sentences in the text describe characters using pronouns. Second, usage of character names not found in the dictionary. Third, insufficient length of the text.

3.4 Methods for Network Analysis

Since the networks we are analyzing in this work are not complex networks covering multiple characters, but rather small networks constructed based on multiple stories, in the process of network analysis, we aim to visually identify whether there are similar structures in different networks. While the structure of visualized graphs is not strictly related to the underlying mathematical calculations of the network, the differences exhibited in the network through the various characteristics of nodes allow for exploratory data analysis through such visual structures (Ahnert et al., 2020). Therefore, we consider that analyzing the morphological features of networks can provide us with another analytical perspective.

Hence, in the following analysis process, we are not only focusing on the morphological features of networks visually, but also calculating features of the network data, such as the number of nodes, features, and network density.

In analyzing the visual characteristics of network structures, we plan to use clustering methods to conduct an overall observation of the shape of the networks. In this process, our aim is to explore what kinds of network structures are most common and what information these common structures can provide us about character relationships. Therefore, we aim to find articles with similar network structures and identify differences in network structures across different stories. To achieve this, we need to analyze a significant number of static images. However, observing images one by one does not allow us to understand the shape of the entire collection and recognize recurring patterns. Hence, we have thought of utilizing image analysis tools to create a categorized "navigation".

Manovich (2012) pointed out that digital image processing can analyze images by calculating visual attribute statistics, such as edge quantities and orientations, which can be used to compare visual differences between different images and for preliminary exploration of large image collections. We plan to initially classify all network graphs using the image clustering analysis feature in the Orange software. This tool categorizes images based on specific features extracted from training images with the assistance of a specific convolutional neural network model (SqueezeNet) (Mohapatra & Swarnkar, 2021). Subsequently, we will evaluate the clustering results and classify them based on network density using the network analysis package in NetworkX. We will analyze and compare these two clustering results, ultimately summarizing the differences in results and information brought about by these two tools due to their different perspectives. We are undertaking this approach because we hope that different comparative methods will provide us with a diverse understanding.

We divided the images into three groups according to the method of constructing the networks. We then sequentially imported them into the software for cluster analysis of the networks under each category, and examined the images based on the branches generated after clustering. Throughout this process, we intend to summarize the common characteristics of the networks in each cluster and compare them based on different clustering branches (Figure 1).

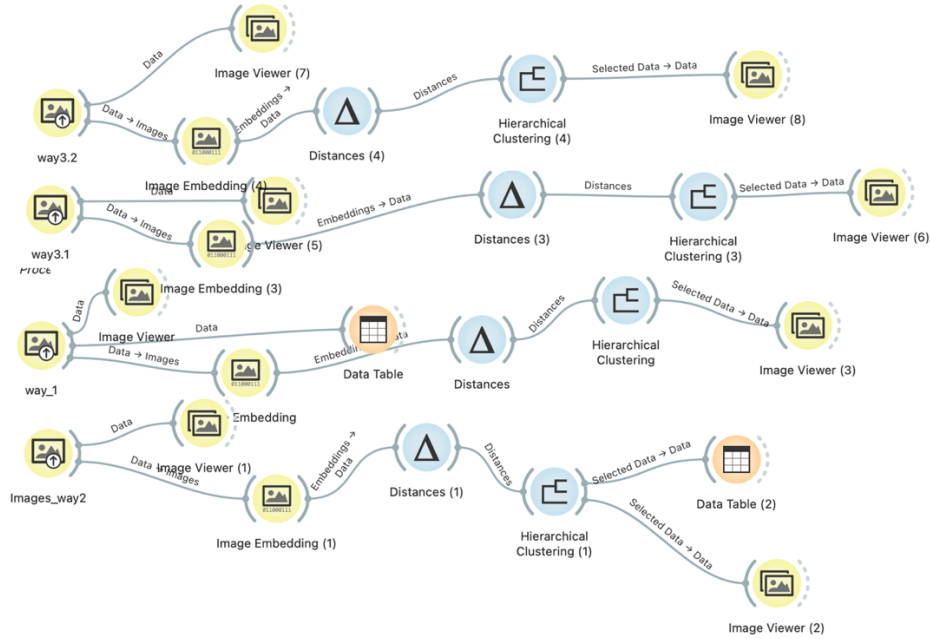


Figure 1: Analyze character networks using the image clustering feature in Orange.

In the process of analyzing networks using visual methods, we focus on several aspects: Firstly, the number of nodes in the network. The number of nodes is an intuitive indicator of the number of characters mentioned in the story. A higher number of nodes in the network may indicate more complex character relationships within the story. Secondly, the number of connections formed by individual nodes and whether the network forms a closed structure. If the connections formed by individual nodes in the network are simple, it may suggest that the story only contains simple character relationships. On the other hand, a closed structure and more connections formed by individual nodes may indicate a certain level of interaction between characters in the story, suggesting closer or more complex character relationships.

In the process of analyzing networks using computational methods, we also focus on the following aspects:

Firstly, the overall structure of the network. When observing the overall structure of the network, we look at it from two perspectives. Firstly, the number of nodes. Because the number of nodes reflects one aspect of network size. By viewing the size of the network, we can analyze how many characters are used in the creation of the story. By examining the network density, we can observe the complexity of the story plot. We assume that when characters appear more frequently, it indicates a more complex story plot compared to a low-density network, suggesting more character relationships or interactions. Next, we observe the shape of the network. In this step, we look at the overall shape of the network, such as whether it forms a completely closed network or if every node is interconnected.

Secondly, the weight of connections in the network. In this step, we observe the weights of connections in different networks because the information conveyed by connection weights varies in different construction methods. By examining the connections, we can observe the extent of relationships or interactions between characters in the respective stories and identify characters that may be designated as core roles or isolated individuals.

Thirdly, node combinations in the network. In this step, we analyze the centrality of nodes in the network. We will analyze this centrality from two common perspectives. Firstly, we look at the degree of nodes, which refers to the number of other nodes directly connected to a particular node. In other words, the degree of a node is its number of direct connections or neighbors. Analyzing the "degree" of nodes in a character network can help us identify key characters in the network because they are directly connected to more nodes. By combining networks constructed using different methods, we can identify important characters in terms of character interactions. Simultaneously, we assume that this method can help us infer whether characters in the story are involved in major plots or themes. If there are fewer connections within the network and isolated nodes are present, we can infer that the character represented by an isolated node is not a key participant in the main plot, perhaps only mentioned. We will validate our assumptions through close reading methods.

4.Data Analysis

4.1 Analysis of the Movie Character Network

We imported the files of nodes and edges from the database into the network visualization software Gephi. We selected the Yifan Hu network layout. In this network, the nodes represent the characters in the movie, and the links are determined based on the time intervals in which two characters appear in the movie. If two characters appear within a short time interval, a link is established between them. The weight of the links reflects the frequency of co-occurrence between the two characters, if the two characters repeatedly appear consecutively within a short timeframe, the weight of the link between them increases accordingly. (figure 2).

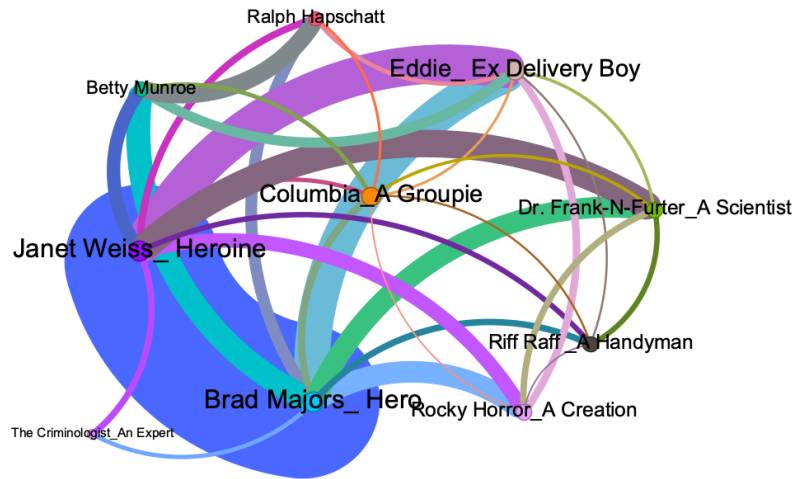


Figure 2: Character network constructed based on the co-occurrence duration of characters in the movie.

From the figure, we can observe the names of the characters, with each node representing a movie character. The links between characters and the weights of links can be determined by observing the thickness of the colored edges in the graph. This network contains a total of 10 nodes, which do not show all the characters in the movie. According to the description of character extraction in the literature, the author has filtered out unnamed or minor characters during the processing of the character list, which indicates that the characters shown are those who appear relatively more frequently. At the same time, we have calculated the network characteristics (as shown in Table 1). There are a total of 32 edges in this network. We calculated the degree centrality of the nodes, as this is one of the metrics used to measure the importance of nodes in the network. Degree centrality measures the number of links an Individual has within a network((Liu et al., 2017). Brad and Janet are the two characters with the highest scores, both scoring 1,

indicating that these two characters' nodes are directly connected to all other nodes.

Also, we calculated the clustering coefficient of the network. This coefficient indicates the probability of forming closed triangles between nodes and their neighbors in the network, which is used to assess the tightness of connections between nodes in a network(Ladd et al., 2017). The value of 0.885 shows a high probability of forming closed triangles among most nodes and their neighbors in the network, suggesting that many characters appear together in the same scenes. Furthermore, we also calculated the betweenness centrality of the nodes. Betweenness centrality measures the extent to which a node acts as a bridge or information transmitter in the network(Liu et al., 2017). In other words, it identifies nodes that connect different parts of the network. The results present that only four characters have calculated values, and the data indicates that Brad and Janet have high betweenness centrality, playing roles as connectors of other characters in the network.

Character name	Degree Centrality	Betweenness Centrality
Brad Majors	1.0	0.1389
Janet Weiss	1.0	0.1389
Eddie	0.8889	0.0417
Rocky Horror	0.6667	0
The Criminologist	0.2222	0
Ralph Hapschatt	0.5556	0
Betty Munroe	0.5556	0
Columbia	0.8889	0.0417
Dr. Frank-N-Furter	0.8889	0
Riff Raff	0.6667	0

Table 1: Calculation results of node characteristics in the network

Based on the weights of the links, we can identify the characters who frequently appear together within a short time frame. Kagan et al. (2020) used a heuristic approach to simulate interactions between characters, assuming that two characters appearing successively within a short period have a certain degree of association. By calculating the co-occurrence frequency, which inferred the relationships between characters. A link with a large weight indicates frequent interactions between two characters, while a link with a small weight indicates infrequent interactions.

In the above figure, we observe that the link weights between Janet and Brad, Eddie, Janet-Brad-Frank, and Janet-Brad-Rocky are the highest. Besides, the link weights within triads such as Brad-Janet-Betty and Brad-Betty-Ralph show significant values. This suggests that, in the film, the characters in the mentioned groups appear in succession within a short period of time more frequently, indicating more frequent interactions among them, playing a significant role in the plot's development. If we infer the closeness of character relationships based on the weights, we can infer that the mentioned character groups maintain close relationships.

4.2 Fanfiction Character s' Network Analysis

4.2.1 Method One: Constructing a Character Network Based on Text Window Co-Occurrences

First, we plotted a histogram of the number of nodes in the network, as the number of

nodes indicates the number of character names mentioned in the story. Observing the chart below (figure 3), we find that the node count is concentrated between 2 and 10, with only a small portion of networks having more than 10 nodes. The peak is concentrated between 4 and 6, which means that most node counts range from 4 to 6. Additionally, we can see that the data skews to the right. This suggests that, in our sample stories, relatively more stories contain between 2 and 6 characters, while it is less common for stories to include a large number of characters (more than 10).

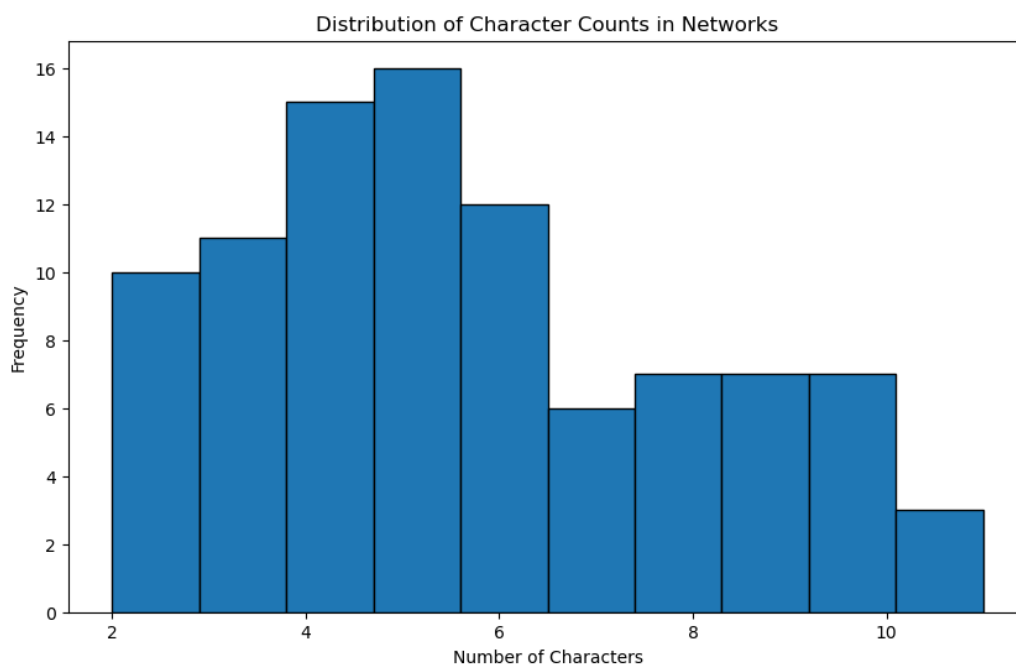


Figure 3: The distribution of the number of nodes composing the network.

Figure above indicates that in the fan fiction stories we selected, a large proportion of the stories are constructed using fewer characters. Of course, we can also understand this as more fan fiction authors wanting to focus only on a specific pair or a few characters to build their stories. Possible explanation for this phenomenon is that, for fans of the movie, only one or a few characters resonate with them or leave a deep impression. Alternatively, they might be dissatisfied with the ending or character relationships of only one or a few characters in the original movie. As a result, they hope to rewrite these characters' settings or continue developing the plots involving their favorite characters by constructing their own stories. These plots can be romantic or conflict-oriented. In the subsequent research, we will continue to explore this phenomenon.

Next, we will examine the structure of the network. First, we will use one of the network results as an illustration to explain what the vertices, edges, and edge weights in the network graph represent.

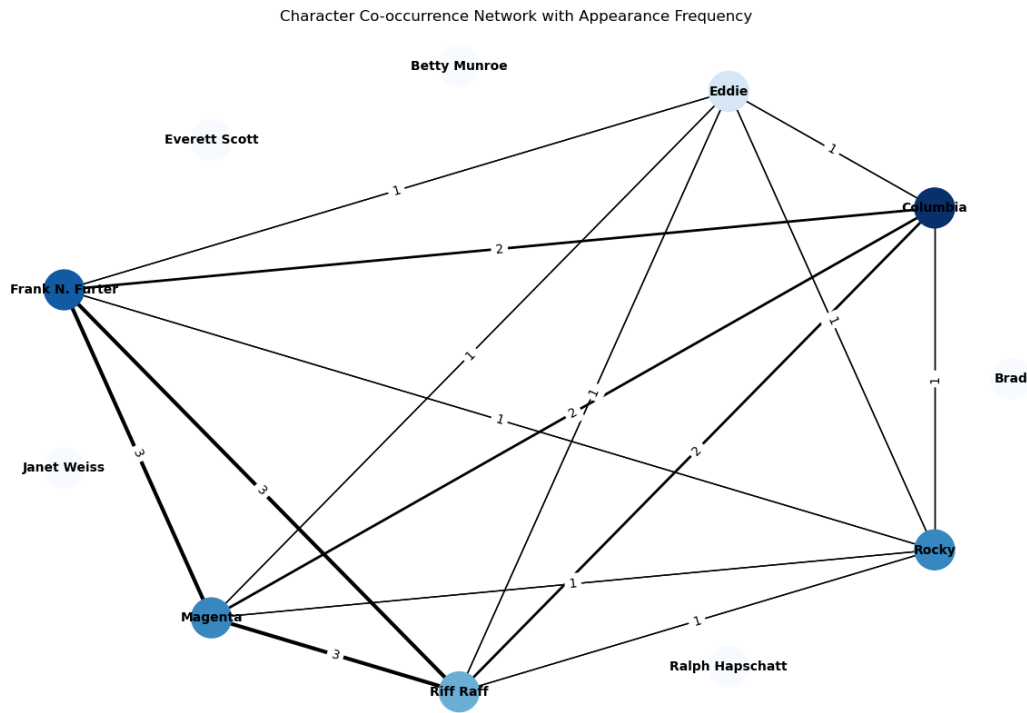


Figure 4: Character network constructed based on the co-occurrence frequency of character names within textual windows.

As shown in the figure, the nodes represent the characters detected in the story, and the color intensity of the nodes indicates the frequency of the character's appearance in the story. Specifically, the darkest nodes represent the most frequently detected character names, such as Columbia in the image. Each link represents the number of times two characters were detected as co-occurring within a textual window. The weight of an edge represents the total co-occurrence frequency of two characters within these windows.

We decided to add a node for each character to the graph, regardless of whether or not they were connected to other characters. This means that even if a character does not co-occur with any other characters in the text, it will still be added to the graph as an isolated node, as in the case of Janet, who was not detected in this particular story. We chose to do this to observe whether characters from the character dictionary appear in the story. This step helps us identify characters that exist but are not mentioned in the story.

Continuing with the above image as our example for analysis: by observing the color intensity of the edges, we can identify character combinations with relatively high co-occurrence frequencies in the network. In the graph, we can clearly observe a prominent triangle—Frank, Magenta, and Riff Raff—which indicates that these three characters frequently appear in the same textual window. Thus, we tentatively conclude that these three characters are the main characters in the story. Compared to other character links, the links between these three are more prominent, meaning frequent co-occurrence implies frequent mention. Therefore, we have reason to believe that these three are the main characters in the story, while other characters might be background characters.

Among the 95 generated networks, we divided them into three clusters. Then, we used image clustering tools for analysis. By using image clustering, we found that 39 networks are of simple structures. Simple means that the number of vertices forming the network mostly ranges from 2 to 4, with only one network containing 5 vertices, and the number

of edges constructing the network ranges from 2 to 6. In 33 networks, the number of vertices significantly increases, ranging from 5 to 8, and the links between vertices also increase significantly, with 28 of these networks forming closed shapes. In 23 networks, the number of nodes ranges from 8 to 11, with even denser links. The simplicity in the structure of the clustered networks may indicate that these stories employ simple character relationships. As the number of network nodes and links increases, it may suggest more intensive interactions among characters within these stories, or that the stories may introduce more complex character relationships with more dense and complex interactions.

Additionally, we identified common patterns of character pairings by analyzing the weights of the links between characters. The following character pairings are more common, with link weights between them all exceeding 1: Frank-Brad-Janet, Magenta-Columbia-Riff, Frank-Brad-Columbia, Frank-Magenta-Columbia, Frank-Janet-Brad, Frank-Brad, and Magenta-Columbia. Frank appears most frequently in all clusters, indicating that Frank is a recurring character and is worth focusing on in our subsequent research, as he seems to attract a lot of attention from readers. Janet and Riff appear in 15 networks each. We also found that the Magenta-Columbia pairing appears multiple times (11 times), with weights mostly being 4 or 3. Next is the Magenta-Riff pairing (11 times) with weights all being 4. Following that is Frank-Columbia, appearing 10 times, all with the highest link weights in the graph. Frank-Brad (15 times), Frank-Magenta (13 times), and Frank-Janet (13 times).

Next, we examine networks classified based on network density. Network density is an indicator of the degree of connectivity within a network. A network density value of 0 indicates no links, whereas a value of 1 indicates a fully connected network. We categorized the networks based on the following results: in the category where network density equals 1, there are only two networks, both forming complete links. In the category where network density is less than 1, there are 27 networks, and 48 networks have densities greater than 0.5 but less than 1.

These results demonstrate that most networks do not form complete links, suggesting that not all characters appear within a certain distance in the text. To identify important characters more accurately within the network, we calculated the degree centrality of the nodes. Calculating degree centrality can reveal which characters have more direct connections within the network.

Upon observation, we found that sometimes multiple characters in the same network exhibit the same degree of centrality. In the process of our analysis, we marked these characters as important individuals. The results are shown in the following figure.

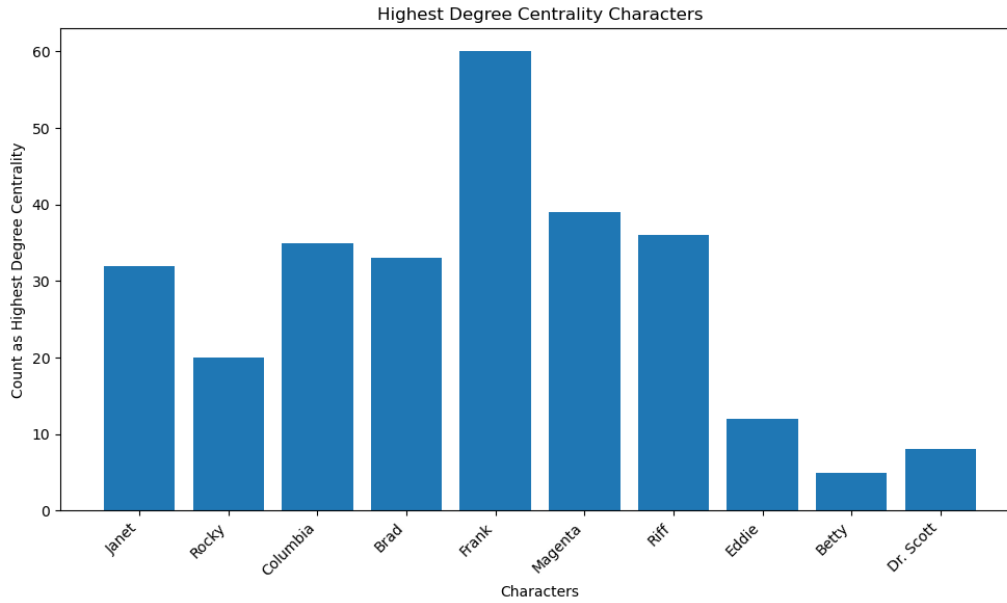


Figure 5: Characters Appearing as Important characters in the Network and Their Frequencies.

In the chart, the x-axis represents character names, and the y-axis denotes the number of times a character appears as a "important character" in the network. We observe that the bar representing Frank is the highest, indicating that Frank appears multiple times across various stories and plays a central role in the social network of these stories. Once again, the data underscores the importance of the character Frank in the minds of fan writers. Given Frank's frequent co-occurrence with other characters in these stories, we have reason to hypothesize that Frank is a primary participant in these narratives. Multiple co-occurrences with different characters within a certain text length suggest that Frank's name is mentioned frequently in the text. Such co-occurrences indicate a high likelihood of Frank establishing connections with more characters across multiple stories. Calculating the degree centrality of character nodes provides us with a unique perspective to understand Frank's significance among fan authors.

Next, we begin to filter out the key character pairings. Using the frequency of co-occurrence of two identical characters in the network as a reference standard, we identify the pairs of nodes with the highest link weights and consider them as key character pairings in a story. We then calculate how many networks also feature this character pairing with higher link weights. The results of our statistics are as follows:

Character Combination	Frequency of Occurrence
Frank-Magenta-Riff Raff	37
Frank-Brad-Janet	40
Frank-Magenta-Columbia	36
Frank-Brad-Columbia	23
Frank-Magenta-Rocky	20

Table 2: Common Character Pairings in networks based on the co-occurrence frequency of character names within textual windows.

The above table reflects the most common character pairings, and these pairings have higher weights, indicating that these characters frequently co-appear within a text window. This suggests that there is interaction between the characters within these pairs. From the above character combinations, Frank still stands out as an important figure within the

group. At the same time, we also notice that Magenta is frequently paired with Frank.

Through above analysis, we can gather some information about character pairings. However, we cannot identify actual interactions or simultaneous appearances of characters within scenes. We consider this to be a limitation of the construction method, as character names co-occurring within a certain text window might simply indicate that they are mentioned together, or by association. After observing the clustering results, we notice that more than half of the stories introduce additional characters and create more links for these characters, (differ from the high-weight connections that appear mainly between Janet and Brad, Eddie, Rocky, and Frank in the film network), which can be interpreted as fans wanting to add more variety to the storyline as they create stories. As more characters are introduced, the plot variations increase accordingly.

4.2.2 Method Two: Constructing a Network Based on Character Dialogues

To further analyze how fans create interactions between characters in their stories, we examined networks constructed based on dialogues between characters. One such network structure is shown in the figure 6.

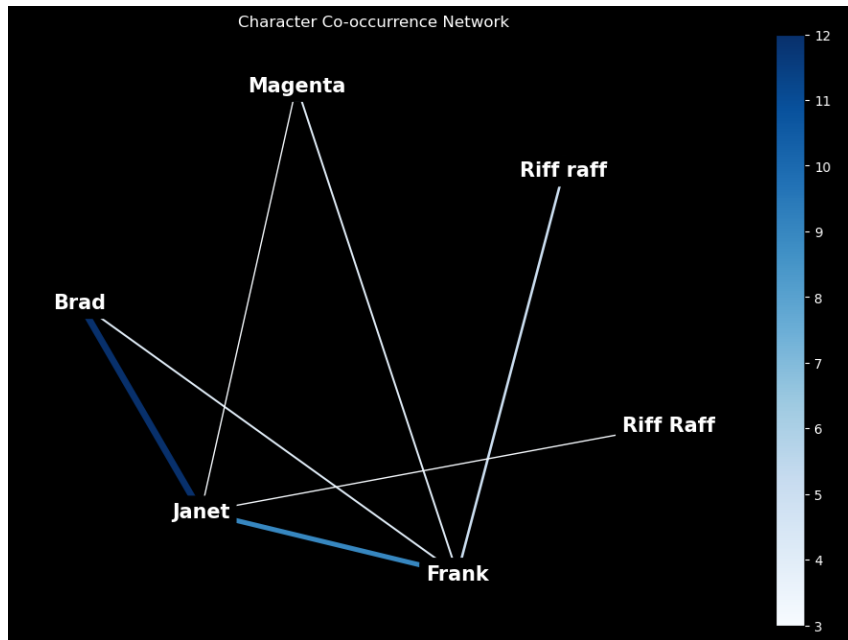


Figure 6: a character network Based on Character Dialogues

In the network, the nodes represent the names of characters in the story, and the connections between them indicate dialogues between two characters. The weight of the links represents the number of dialogue sentences between them. Through observing such a network, we can identify character pairings that frequently interact in the story and the total number of dialogue instances. We regard dialogue as a type of interaction, which allows us to recognize the frequent character pairings and the cumulative frequency of their dialogues., thus identifying the main participants in the story.

Firstly, we plot the number of nodes in the network as a histogram. By noting the histogram, we can compare the number of characters engaging in dialogues across different stories. This helps us understand the complexity of the story constructed by the

author. Generally, more characters participating in dialogues indicate more direct interactions between characters, suggesting higher complexity in the story's interactions.

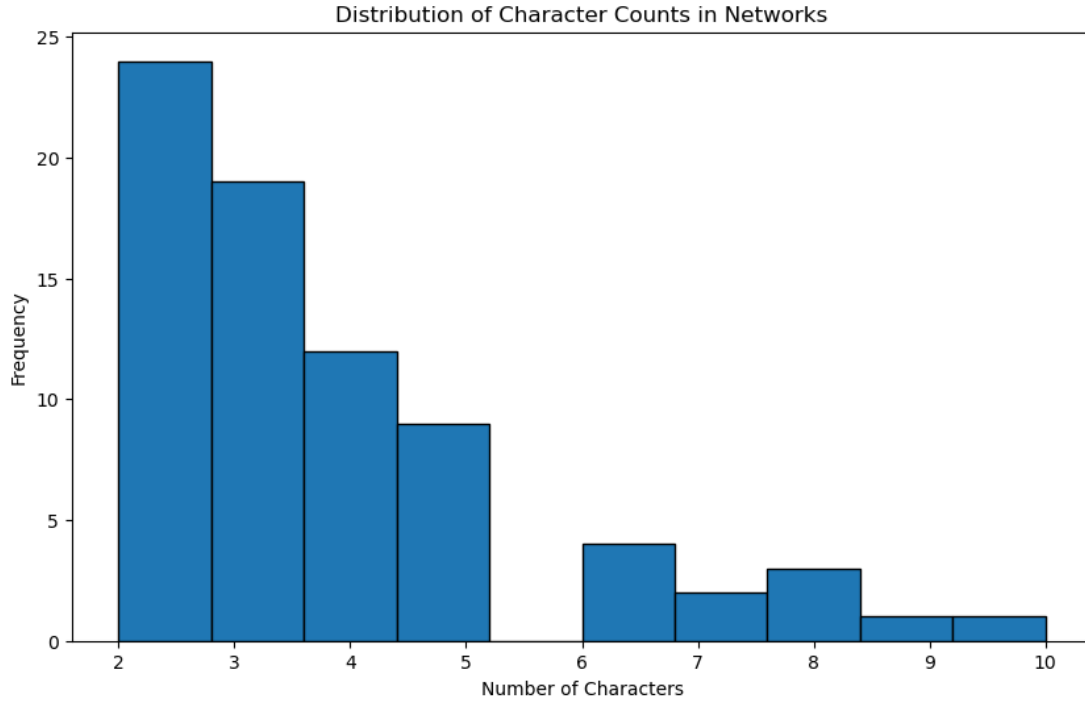


Figure 7: Histogram of Node Count Distribution

In the figure, the horizontal axis represents the number of nodes, and the vertical axis indicates the number of networks within each node count interval. By observing the distribution of node counts on the horizontal axis, we can see that there are significantly more networks within a certain node count range. Specifically, most networks have a node count concentrated between 2 and 5. Examining the distribution on both ends, we can determine that a few networks have very few (2) or very many (10) nodes.

Most of the bars are concentrated within the smaller node count range, with the 2 to 5 node count range being the most common. This indicates that most networks have a smaller number of nodes, suggesting that most stories involve a relatively small number of characters engaging in dialogues. Conversely, the few networks with more nodes likely reflect more complex character interactions.

After using a clustering tool to analyze the networks, we could not observe any useful similarities within the clusters for our study, and thus, this will not be discussed further.

Next, we clustered the networks based on the results of network density. Here, network density reflects the story. There are five networks with a density of less than 0.5. A total of 33 networks have a network density of 1, characterized by all vertices being interconnected. Among these, some networks form closed triangles, polygons, or lines between two vertices. In the category of networks with density ≥ 0.5 , there are 37 networks, all of which share the common feature of not forming closed structures.

From the above results, it is evident that most networks have a high network density. Networks with a density of 1 indicate that each character interacts with every other character, meaning that characters arranged with dialogue interactions in the story also

engage in dialogues. Besides closed structures, we also discovered "star-shaped" structures, as shown in the figure (figure 8).

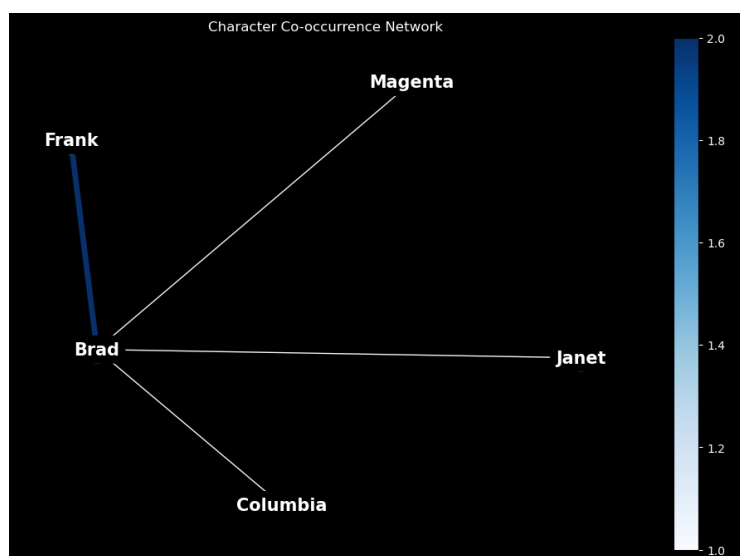


Figure 8: A network with "star-shaped" structure

The above structure shows that in this story, Brad is the "information hub" because he is the character who interacts with the most other characters in the diagram.

At the same time, by analyzing the weights of the connections, we can identify triangular interactions, which can help us recognize common dialogue interaction pairings. (see figure 9).

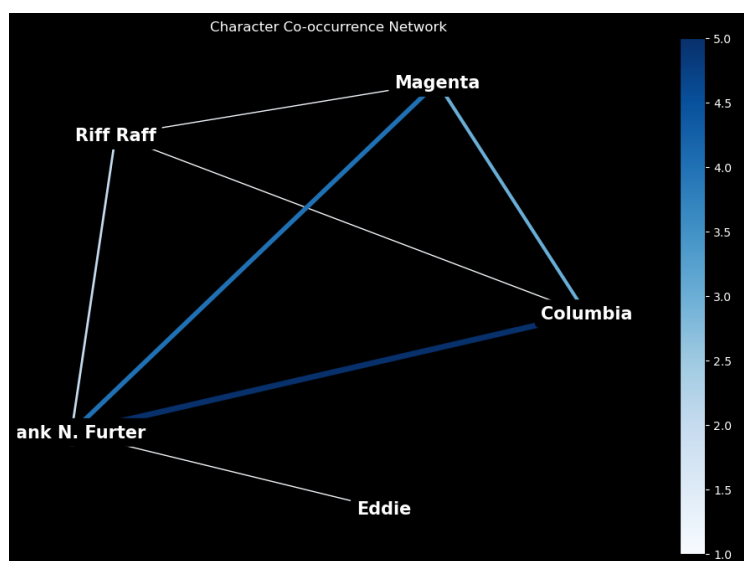


Figure 9: Based on the color of the connection weights in the diagram, we can identify triangular character relationships with frequent dialogue interactions.

In the above diagram, we can determine the three characters who most frequently interact based on the depth of the connection colors, and the characters between whom interactions also occur. This allows us to identify character pairings that frequently engage in dialogues according to the illustration.

Next, we identify common character pairings. Our method involves observing the colors of the edges in the network diagram and recording the nodes linked by the darker edges, as darker edges indicate higher weights, implying more frequent interactions between

those two characters compared to others in the story.

Character combination	Count
Magenta-Riff	7
Magenta-Columbia	13
Columbia-Frank	6
Frank-Brad	11
Janet-Brad	13
Janet-Frank	8
Frank-Magenta	10

Table3: Common character combination

Based on the table above, we can observe that a significant number of stories feature frequent dialogues among the character pairings of Magenta-Columbia, Frank-Brad, and Janet-Brad. From this, we infer that these characters share close relationships within the stories. However, we cannot determine the nature of these relationships or the themes of their dialogues. Nevertheless, in terms of character arrangement, Frank, Magenta, Brad, and Janet are frequently chosen and arranged to engage in dialogues in more stories.

Additionally, we are interested in identifying the "hub" of the network, the character who interacts with many others. To this end, we calculated the degree of each node. The degree of a node is the sum of its connections. In networks, the node with the highest degree is the one who interacts with the most characters. The statistical results are illustrated in the figure below.

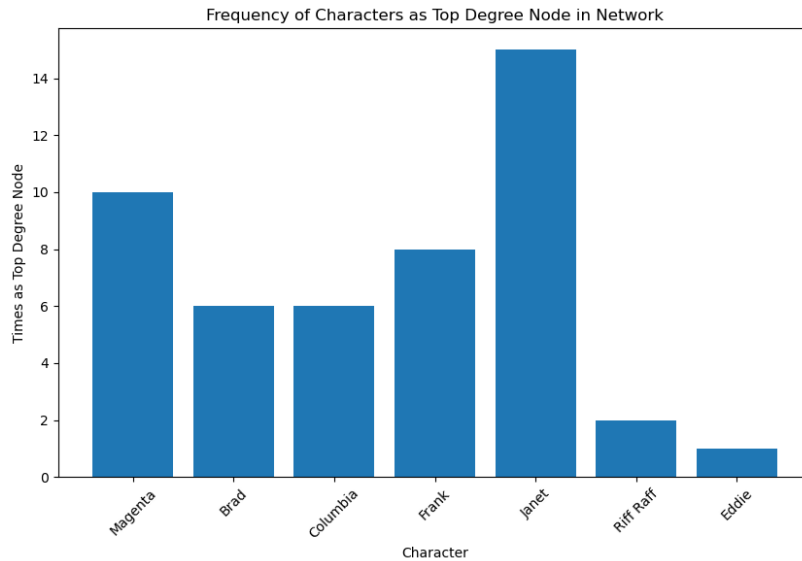


Figure 7: Frequency of Characters as top degree node in character network

In the figure, the x-axis represents character names, and the y-axis indicates the frequency of each character being the most connected node in the network. It is evident from the diagram that Janet is the character with the highest frequency. Unlike the network analysis results constructed using the first method, in the dialogue network, Janet can be regarded as an influential character, as she engages in conversations with the most people in 15 stories.

When analyzing dialogue networks, we can obtain information about character interactions, such as who participated in conversations and who talked more with whom. This allows us to infer close relationships between characters. However, we can gather very little information about the nature of these relationships because high dialogue interactions can occur whether the characters are enemies or lovers.

4.2.3 Method Three: Constructing Networks Based on Verb Classification

Before proceeding with the analysis, we select one network image (as shown in the figure) as an example to explain what the nodes and links in a network represent.

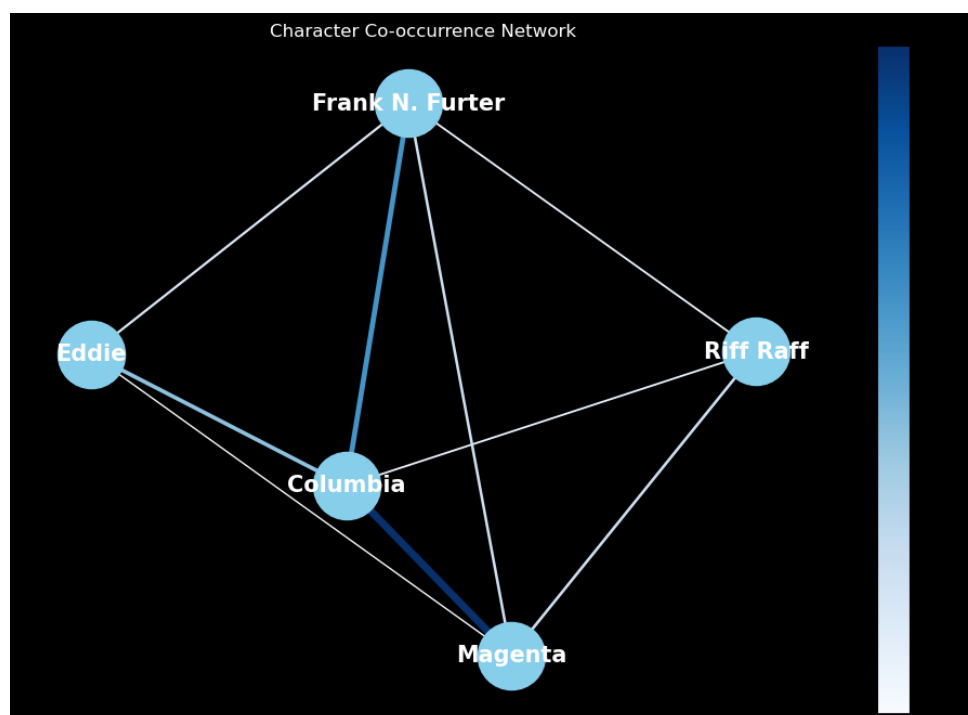


Figure 8: A character network based on Verb Classification

In the network, nodes represent characters from the article. Each link represents two characters that not only appear in the same sentence but also have the sentence's primary verb listed in a predefined dictionary. The weight of the links is accumulated from the sentence scores, which are calculated based on a verb-score dictionary. Therefore, each link can be viewed as an interaction between two characters. Through examining the frequency of the same character nodes across all networks, we can identify the main characters used by fans to create stories.

By examining the links between nodes, we can determine which characters interact with each other. Higher link weights indicate more frequent interactions or interactions involving higher-scoring verbs. Furthermore, we can identify primary and secondary characters based on link weights. For instance, in a network comprising six nodes, if the links among three of the nodes have higher weights compared to the links among the other three, we can conclude that the group with higher weights constitutes the main characters. This is because they exhibit more intense interactions, indicating their relationships are relatively more important or closer.

Apart from that, we have summarized the number of nodes from 91 networks and created the following histogram. The x-axis represents the number of nodes, and the y-axis represents the number of networks corresponding to that node count.

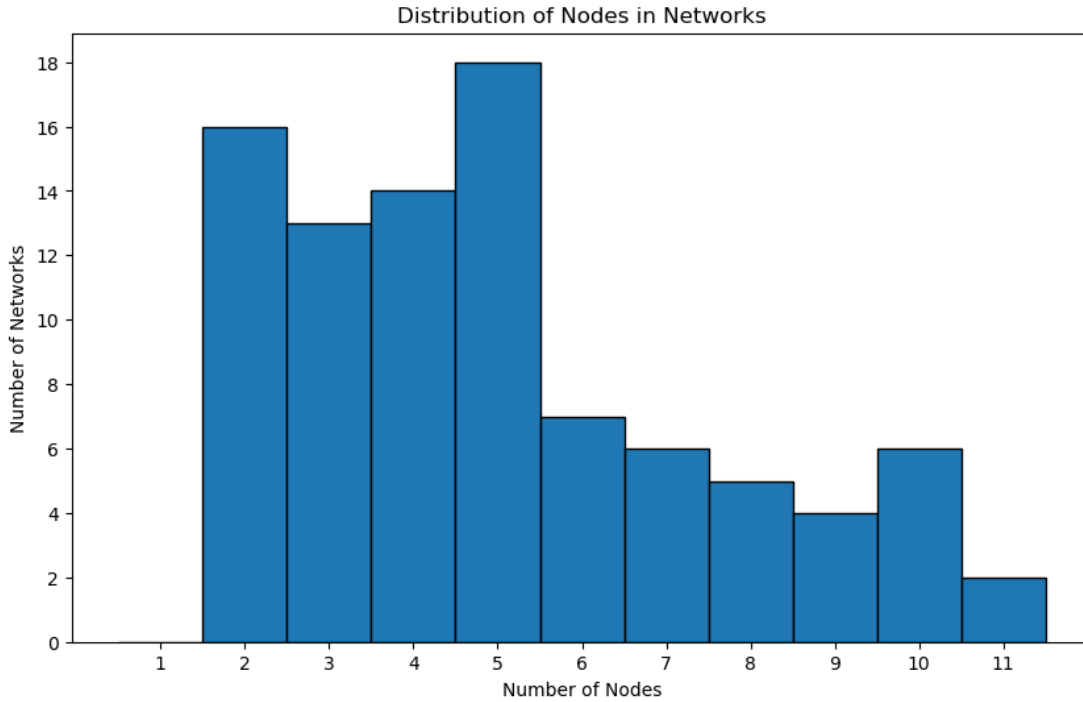


Figure 9: Distribution of nodes in Networks

We can observe that the chart displays a single peak corresponding to a node count of 5, indicating that most networks consist of 5 nodes. Other nodes with higher frequencies include 2, 3, and 4, which reflect the characters involved in actual interactions or actions within the stories. This observation suggests that the majority of story networks are relatively small in scale. However, the number of nodes is only one dimension in determining network size, further analysis integrating network structure is needed.

We also employed image clustering methods for analysis. Through image clustering analysis, through this analysis, we identified various configurations within the networks, including two communities, community-linked isolated points, and linear structures. We observed that within the nodes constituting these communities, Frank serves as a central member in each community, frequently forming clusters with Riff (4), Magenta (5), Brad (4), Columbia (5), and Janet (5)—the numbers indicate the frequency of connections formed with Frank. Conversely, Betty (6) and Eddie (8) were the characters most frequently appearing as isolated points—the numbers indicate their occurrences as isolated points. These patterns indicate the differences in the combinations of interactions among characters.

Furthermore, we discovered central characters within some communities. These characters are pivotal in the networks, either acting as intermediaries linking communities and isolated points or being central in a linear structure. For instance, Magenta connects Riff and Columbia as an intermediary, or Janet acts as a central figure in a community linked to the isolated node of Frank. Besides, we observed triangular closure structures, with the most common triangle relationship being among Frank, Brad, and Janet, occurring a total of 14 times.

After categorizing the networks based on network density, we re-evaluated the clustering results. In the classification with a network density of 1, there are 31 networks. These networks exhibit a closed structure, where links form between two nodes or create triangular closures. In the classification with a network density greater than 0.5 but less than 1, there are 43 networks. There are 14 networks with a density less than 0.5, which predominantly feature two unconnected communities.

This network density-based classification method provides us with a perspective that combines network density with visual structural characteristics of the networks, offering us insights into network structural features. The classification results for high network density indicate tighter connections between nodes, forming denser communities or closure structures. Conversely, lower network density may reflect sparser connections between nodes, resulting in a more dispersed network structure with two unconnected communities. However, so far, we have not been able to identify more effective information regarding interactions between characters from the above analysis.

Next, we focus on character combinations and identify common character combinations within the network. Our identification steps are as follows: first, we examine the darkest links in the network one by one, recording the two or three characters connected by each link. We regard such character combinations as the main character combinations within a network. Finally, we count the frequency of each character combination appearing as a "main character combination". Also, we will observe the character combinations identified from the previous two types of networks (text window character network and dialogue character network) to check for overlaps with the combinations identified by the above two methods. We will present the results of the main character combinations in the following table.

Character combination	counts
Frank-Magenta-Columbia	3
Frank-Magenta-Riff Raff	8
Frank-Brad-Janet	9
Janet-Brad	10
Magenta-Columbia	9
Brad-Frank	13
Frank-Riff Raff	13

Table 4: Common Character combination

Based on the above figure, we have discovered new character combinations, such as Frank-Riff, indicating that this combination is more prominent or important across multiple networks. Additionally, we have observed character combinations commonly found in the previous two types of networks, such as Frank-Janet and Frank-Brad.

This observation raises our curiosity about whether, in such cases, the central nodes in the networks are the same as the previously identified "most popular character" Frank and "most sociable character" Janet. Therefore, we calculated the degree centrality of nodes in each network. We identified the characters with the highest degree centrality, excluding networks with only two nodes. In cases where multiple characters had the same degree centrality, we included all of them in our count. The histogram below shows the calculation results.

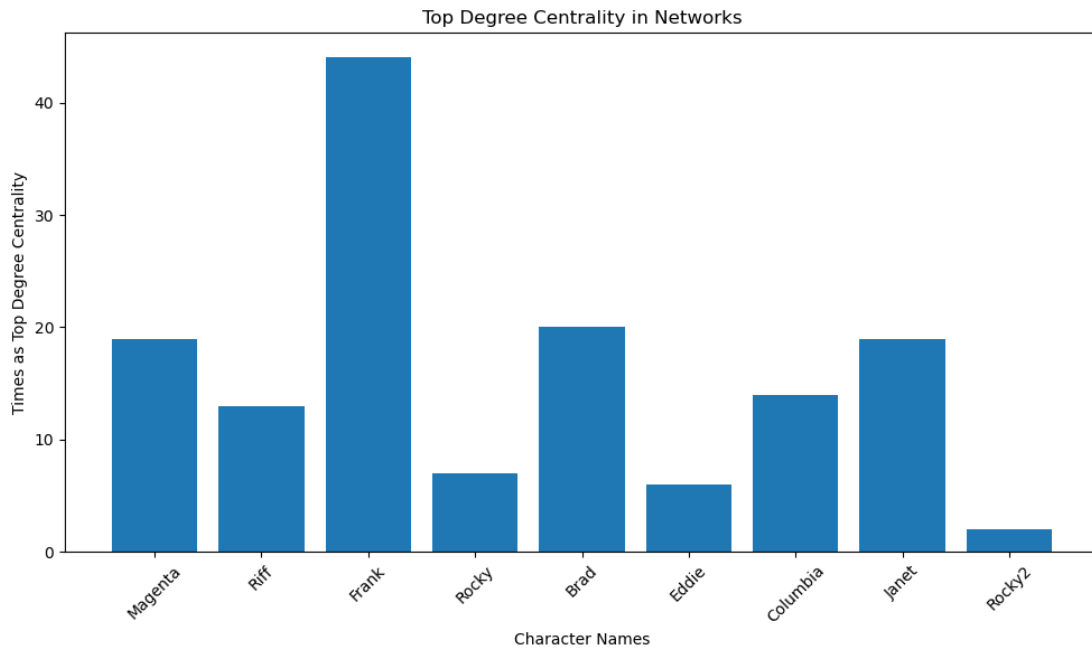


Figure 10: The frequency of characters with highest degree centrality

In the chart, the x-axis represents character names, and the y-axis represents the number of times each character has the highest degree centrality in the network. It is clear that Frank stands out as the leading character in this dimension. This means that Frank is the character who has interactions with the most other characters. This high degree of centrality can be interpreted as an indication that this character plays a crucial role in the entire story. When the same character repeatedly shows high degree centrality in the network, we can infer that fans consider this character to be a main character when constructing their stories. Indeed, in the original movie, Frank is one of the main protagonists, suggesting that fans assign considerable importance to Frank in their creative stories.

Of course, we also want to further explore the nature of the interactions between characters. Therefore, we analyzed networks constructed based on the classification of central verbs in sentences. Each image corresponds to a story (as shown in the figure 11). Before the analysis, we first explain the nodes, edges, and edge weights in the network.

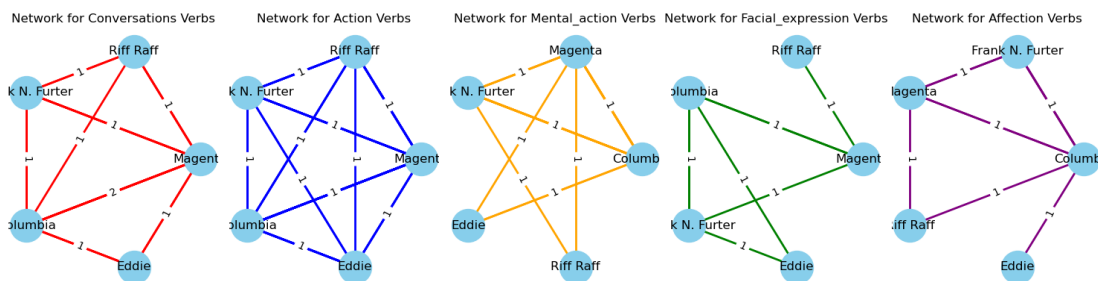


Figure 11: The characters' network based on the classification of verb types

In this figure, we can observe combinations of five different network shapes. This is because we classified the main verbs identified in sentence structures into five categories. Therefore, if five networks are displayed in the image, it indicates that the story content

involves these five types of interactions, with certain characters participating in these interactions. Some images may only show three or two networks, suggesting that the story lacks depictions of certain verb-type interactions.

By analyzing the network density of such charts, we can understand which types of interactions the author focuses on describing in a story. Observing the connections between nodes in the network, we can identify the diversity of interaction types between characters. For example, under the verb category, we may only see a network connection between two characters, whereas, in expressions of affection, we can see a network connection involving six characters. In the verb category, only Janet and Brad form a connection, while in the affection category, besides Janet and Brad, Magenta and Columbia also form a connection. This not only illustrates the different types of interactions among these characters but also highlights the number of characters involved in different types of interactions, which is the diversity of character interactions we aim to explore.

Of course, identifying interaction types in a story using this method is still influenced by the issue of referencing, as we cannot accurately identify references that do not pertain to specific characters. Analyzing this network allows us to observe the types of interactions more intuitively between characters. Compared to the previous network, we can clearly see how the author constructs interactions in the story, how many characters participate or execute these interactions, and we can also compare the network density constructed from different stories to observe the author's preference for constructing certain interactions.

Overall, these networks take on various shapes, including closed hexagons, triangles, and squares, as well as networks that do not form closed shapes. We calculated the number of nodes and the network density for different verb categories and used this data to determine whether the network structures for certain verb categories are more complex than others in different stories. Specifically, we compared the network density and the number of nodes in all networks within a single image. First, we compare the number of nodes, if the node counts are the same, we then consider the network density values. This is because network density helps measure the compactness of a network and can partially reflect the intensity of character interactions. A high-density network might indicate that this type of verb is used more frequently in the story.

During our analysis, we will consider equivalent cases. For example, within a single image, if two verb category networks have the same number of nodes and network density, we include both verb types. However, if all five networks in an image have identical feature data, we will not consider such a case. The statistical results are shown in the chart below:(figure 12)

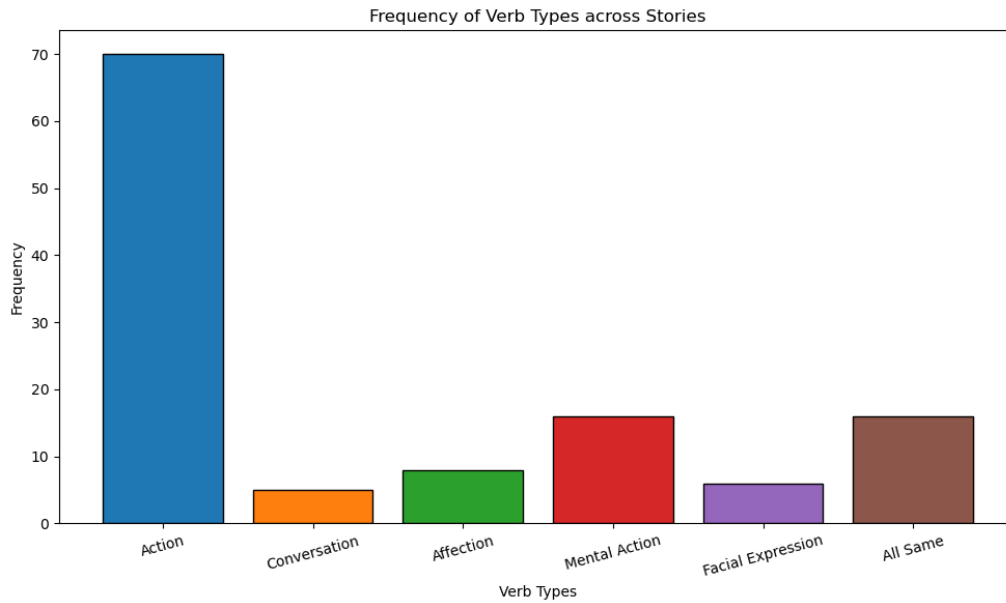


Figure 12: Distribution of high network density under specific verb types

By observing the above chart, we can see that data indicate higher network density under the action category in 70 out of the 91 stories. Referring to our previous description of the "Action" category verbs: "Verbs marked as action are those expressing daily activities and behaviors, including words representing everyday actions and expressions of emotions other than affection, such as move and find." This suggests that specific action verbs may frequently appear in most stories, indicating more frequent physical interactions among characters. In contrast, only five stories show higher network density for dialogue verbs, implying that fewer stories rely more on verbal communication to describe character interactions.

The differences in the number of stories under various verb categories provide us with some insights. First, it reflects different forms of interaction. The use of these verbs may involve actual actions and events between characters, and high-density networks indicate that the authors' stories emphasize action descriptions more than direct dialogue occurrences. In addition, we can infer that stories with high-density networks in the psychological activity category might tend to describe the inner world of characters, focusing more on emotional descriptions.

As mentioned earlier, the most frequently occurring central word in sentences is "smile," a verb expressing affection. Based on this, our next analysis will first explore which characters constitute the network under affection-related verbs and observe the weight of connections between character combinations. Previously, we noted some common character combinations in the dialogue network and verb score network. Here, we focus on whether these character combinations frequently appear under specific verb categories. We will first observe which character combinations appear under different verb categories and whether their connection weights are higher than those of other characters under specific verb categories. Then, we will count the results of the character combinations.

Our process of identifying character combinations is as follows: given the variety of character combinations forming the network, we needed some character combinations as

guides for our analysis. Thus, we chose the most representative character combinations, which include the central characters and common combinations identified in the above study. The combinations are Frank-Magenta-Columbia, Frank-Magenta-Riff Raff, and Frank-Brad-Janet. The statistical results are as follows:

Character combination	count
Magenta-Columbia	16
Brad-Janet	14
Janet-Frank	12
Frank-Magenta	10
Frank-Brad	11
Frank-Riff raff	6
Magenta-Riff raff	13
Frank-Columbia	8
Magenta-Rocky	2
Frank-Rocky	7
Janet-Rocky	4
Eddie-Columbia	6

Table 5: Common character combinations base on connection weight

After viewing the above chart, we can identify different character combinations with high-weight connections under the affection-related verb category. The top three character combinations are Magenta-Columbia, Brad-Janet, and Magenta-Riff Raff. From this table, we infer that in a certain number of stories, fanfiction authors chose to depict interactions expressing affection between Magenta and Columbia, two female characters from the movie. To confirm this, we revisited our initial filtering of AO3 tags because, on AO3, if a relationship tag is in the form of "___/___," it indicates a romantic relationship between the two characters listed under the tag in the text.

To further validate the above result, we examined whether stories corresponding to networks where Magenta and Columbia display high-weight connections under the affection-related verb category were tagged with romantic relationships and conveyed romantic relationships in their content. We checked the corresponding networks of 20 stories under this tag and found that only 12 networks showed a weighted connection between Magenta and Columbia. We employed the same method to examine stories under the Brad-Janet tag, only 15 out of 20 stories demonstrated a connection between the two characters under the corresponding verb category. Similarly, for the Magenta-Riff Raff tag, only 10 out of 20 stories displayed a corresponding network connection.

Based on these results, we can see that inferring the depiction of character relationships in stories merely from character connections and connection weights in the verb-type networks has limitations and potential biases. However, we cannot deny that identifying romantic character combinations through character networks under specific verb types provides valuable insights.

Firstly, we can observe adaptations in the types of character relationships. For example, in the film, the relationship between Magenta and Columbia is not extensively developed. We only see that both serve as Frank's assistants in the castle, and they act as plot drivers in specific scenes. Columbia displays affection for Eddie, laying the groundwork for later revealing Eddie's death, while Magenta conspires with Riff, hoping to overthrow Frank's

rule and setting the stage for Frank's demise. However, in fan fiction, authors choose to focus on these two characters and develop a romantic relationship between them. Similarly, the film does not reveal much about the relationship between Magenta and Riff Raff. In scenes where they appear alone, there are no hints of a romantic relationship. But compared to other supporting characters such as Betty, they play a more significant role in advancing the plot, such as greeting Jante-Brad at the castle, plotting to release Rocky, and ultimately betraying Frank towards the end of the film. In fan stories, these two characters become the protagonists of a romantic narrative.

Secondly, we discovered the diversity in interactions of the same character relationships and the varying frequency of these interactions in fanfiction, enriching the possibilities of character combinations.

If the high-weight connections identified in the verb-type networks are considered as indicators of close relationships between characters, it is evident that the authors are reshaping the relationships envisioned in the original work. For example, in the movie's character network, the connection weight between Janet and Eddie is higher than that between Eddie and Frank. However, in the listed character combinations, we do not observe a pairing between Janet and Eddie. Similarly, although Brad and Eddie demonstrate high interaction in the movie network, they do not appear in the list of combinations. On the contrary, while Rocky and Janet have weaker connections in the movie network, they are listed in the combinations. Additionally, Columbia, who has weaker connections in the movie network, frequently appears in the fanfiction networks and forms high-weight pairings with multiple characters in different verb-type networks. Not all secondary characters gain the favor of fanfiction authors, some, such as Betty and Dr. Scott, appear in the movie network but are rarely seen in the fanfiction networks. Although this data cannot precisely reveal how fanfiction authors handle different character relationships, we can observe the diversity in fanfiction authors' creative processes from the frequency changes in character combinations across different verb-type networks.

5. Conclusion and Discussion

This chapter aims to summarize and explain the research findings, as well as to discuss our analytical results in conjunction with relevant theories. On the top of that, it explores the methods used for constructing and analyzing character networks, addresses the limitations encountered during the research process, and proposes ideas for future studies on fanfiction.

5.1 Summary of Research Findings

In this section, we provide a preliminary summary of the above analysis results by answering the sub-questions, and then we address the main research question.

Sub-question 1: Which characters in the original film have the highest interaction frequency, and what are the most common character pairings?

By examining the film's character network, we find that the characters Janet and Brad have the highest interaction weight. These two characters are not only the protagonists of the film, the story is also told from their perspectives, and their relationship is established as a married couple. We identify the most common character pairings based on the weights of the links between characters. Janet and Brad, other pairings such as Janet and Eddie, Brad and Betty, Janet and Frank, Brad and Frank, Brad and Eddie, and Brad and Rocky show significant interaction weights. Analyzing these character networks in the context of the film's plot, the following relevant information can be reflected.

The relationship between Janet and Eddie is not explicitly depicted in the movie, the higher-weighted link between them might be due to the narrative necessity of having them appear in the same scene. For instance, there is a scene that explores the relationship between Frank and Eddie. Because Janet and Brad are the main perspectives, the film frequently switches between their reactions, resulting in a short time interval between their appearances and Eddie's. This explains the higher-weighted connections between Eddie and both Janet and Brad in the film character network.

In contrast, Frank and Eddie, who are actually related characters according to the storyline, show a lower weight in the graph. This phenomenon could be attributed to camera transitions, although they are participants in the same plot, the frequency of their short-time successive appearances is relatively low due to the film's editing. In the movie, Frank is a major character, while Janet is a character attracted to him. As two key figures in the narrative development, the high-weighted link between them in the character network suggests multiple interactions or associations. The frequent co-occurrence and interaction are significant indicators of a close relationship between the characters. Also, there is a high-weighted link between Brad and Frank. According to the movie, Brad is attracted to Frank.

To some extent, the high-weighted connection as shown in the film's character network reflects the close relationship between the characters, but about the real character relationship information, we still need to follow the content in the movie.

Sub-question 2: Are there differences in character pairings between fanfiction and the original film? If so, what are they?

We found that certain character relationships from the movie were carried over into the fanfiction. The pairing of Brad and Janet consistently appeared with high frequency across all types of networks based on construction methods. In addition, the combinations of Brad-Frank and Janet-Frank also showed a high occurrence rate. The continuation of these character relationships suggests that these relationships in the film resonated with fanfiction authors. The analysis results show that most stories-built connections and interactions around these three characters, with some authors choosing to further explore or extend the romantic relationships between them. In the network constructed based on the 'affection' category of verbs, these character pairings frequently appeared with strong connections.

We also identified some character combinations that did not appear in the movie character network but did appear in the fanfiction character network. The network based on character co-occurrence within text windows revealed that Frank-Magenta-Riff Raff, Frank-Brad-Janet, and Frank-Magenta-Columbia were common character groupings, indicating that these characters frequently appeared together within the same text windows. In the analysis based on character dialogue interactions, we identified the frequent dialogue pairings of Magenta-Columbia, Frank-Brad, and Janet-Brad. Magenta did not prominently appear in the movie character network, and thus, the aforementioned connections were absent. However, in the movie, Magenta is also a character who advances the plot, this discrepancy could be due to an error in the data from the movie character network we used.

Besides, a certain number of fanfiction authors chose to construct romantic relationships between new character combinations, as observed through the relationship tags in the collected story data. By examining the network based on the "affection" category of verbs, we found that character pairings like Magenta and Columbia frequently appeared with high-weighted connections in the network. We infer that this result shows some authors used these characters in their stories and built romantic relationships between them. This inference was confirmed by checking the character relationship tags, which aligned with the network analysis results.

Sub-question 3: What are the most common interaction patterns among these character pairings?

After analyzing the character network constructed based on verb categories, we found that more stories exhibited higher network density in the general action verb category. This suggests that specific action verbs, such as walk, search, try, and nod, frequently appear in most stories. In contrast, fewer stories showed high network density in the dialogue type category, indicating that fanfiction authors might focus more on action descriptions rather than explicitly highlighting dialogues.

Main Research Question: What is the difference in terms of creating character networks between original movies and fanfiction?

Firstly, in the character networks of fan fiction, significant character relationships are preserved. In the film's character network, the links between Brad, Janet, and Frank have high weights. In fan fiction's character networks, the link patterns involving Brad and

Janet appear with high frequency across various networks, and Frank is an important node in different types of networks. This preservation demonstrates that fan fiction authors do not entirely overturn the original narrative settings when creating their stories. Instead, they use content from the original works as plot clues for their stories. Secondly, relationship pairs like Brad-Frank and Janet-Frank frequently appear with high-weight links in various types of networks, especially appearing with high weights multiple times in “affection” verb-type networks. Referring to the film's content, these two sets of characters are involved in key plot points, where Frank first seduces Janet and then seduces Brad. This also suggests the focus and response to significant plot developments in fan fiction, demonstrating support for these key narrative events.

Besides, in fanfiction networks, more secondary characters are given prominence. For instance, connections between Columbia and Magenta do not appear in the film's network but are found in fanfiction networks but Magenta, Columbia, and Riff commonly form networks in fanfiction. The analysis of networks constructed based on verb types shows that Magenta and Columbia are often placed in romantic relationships in the stories, suggesting that authors use fanfiction to supplement character relationships that are not explored in the original work.

5.2 Discussion on the Adaptation Methods of Fanfiction Based on Network Analysis Results.

In this section, we will further integrate our analysis results with known conclusions from the literature review to discuss the strategies for adapting canonical characters in fanfiction.

In the fanfiction stories analyzed, the main character from the film, Frank, remains a central figure. In the character co-occurrence network constructed based on text windows, Frank frequently appears as an “important character”, the results of the dialogue network analysis indicate that the character Frank not only has a substantial amount of language description but also engages in dialogue with more characters. In the character network constructed based on verb types, Frank is frequently at the center of the community and appears in character combinations within closed triangular networks. Apart from that, in the dialogue interaction network, 'Janet' continues to serve as a key connecting character.

Based on the observations mentioned above, we believe that this reinforces the known conclusions in Brotrager et al. (2023, p.650), which suggest that the transformation from the original work to fanfiction can be explained using the concept of constrained aesthetics: classic characters and their established traits and characteristics are not merely copied and pasted into new fan narratives but are appropriated and utilized. In the film, Dr. Frank-N-Furter (played by Tim Curry, dressed in black garters and stockings with heavy, seductive makeup) is the central figure and the protagonist of the story. Also, in fanfiction stories, this character is given significant attention by authors. By conducting a close reading, we can further supplement the results of network analysis, confirming the protagonist's importance in fanfiction stories, especially by retaining the character's portrayal from the film. For example, the following passage describes Frank's appearance in a fanfiction: “She gripped the banister as Frank, in all his splendor, came toward her. She took him in from eyeliner and lipstick, to pearls, to black semi-transparent peignoir, to feathered mules, and quietly forgot to breathe.” In the film, Frank also makes an entrance with heavy makeup, wearing a black dress, and moving with a model-like stride.

Using a close reading approach, we randomly selected ten fanfiction stories featuring Frank as the protagonist. We found that fanfiction often re-focuses on central characters from the original work, partly filling in gaps left by the original narrative (Lee, 2011). For instance, story #49454569 expands in detail on a romantic scene focusing on Frank from the film.

Additionally, the narrative perspective in fanfiction shifts from the original film but remains the main characters. In the film, we follow the perspective of Brad and Janet, whereas in fanfiction, the narrative perspective becomes more diverse, e.g., story #52511266, which places Frank and Janet in different settings to portray their happy scenes, highlighting their romantic relationship, and story #37372630, focusing on the romantic relationship between Frank and Brad, whereas in the film, Frank has romantic interactions with both Janet and Brad). As our results indicate, fanfiction authors are not entirely overhauling the original narrative setup but are also improvising the content from the original work as plot cues for their fan stories (Jenkins, 2012; Bond & Michelson, 2003).

In addition, certain supporting characters in the film more frequently become protagonists in fanfiction, with stories unfolding from their perspectives. The analysis results using the first type of network construction method show that among the characters identified as important in the story, Frank had the highest number, followed by Magenta, Columbia, and Riff. The statistics for these three characters even surpassed those of Janet and Brad. This outcome is also consistent with Brottrager et al. (2023), which found that characters who were not as direct or prominent in the original universe became important and were brought to the forefront, with characters absent in the main narrative reappearing.

Another example is from the analysis of dialogue networks and verb networks, where we found that Magenta, Columbia, and Riff frequently form parts of the networks. In the film, Magenta, Columbia, and Riff are Frank's assistants and appear as supporting characters. Their actions are typically captured to propel the plot, such as Riff deliberately letting Rocky escape to meet Janet. The film provides very little background and character relationship descriptions for them. These analytical results respond to the findings of Lanier and Schau (2007), which noted that fanfiction often enriches narratives by expanding on minor characters from the original work.

The treatment of secondary characters in fanfiction has been further verified through close reading. For example, story #54502897 narrates the story between Magenta, Columbia, and Riff from a third-person perspective, where Frank, originally the protagonist, is merely a supporting character. In story #53459851, Riff is the main character, and the story extensively describes Riff's psychological activities to tell his story. These stories underscore the greater emphasis on and more multi-dimensional descriptions of secondary characters in fanfiction. Story #1060979 focuses on the backstory of Riff and Magenta, but the background description serves to create a new story setting, it creates a new story setting where they are depicted as brother and sister. In story #51255037, Magenta is no longer a villain alongside Riff, instead, she becomes a compassionate character who ultimately betrays Riff to stop his evil.

These findings illustrate that fanfiction often takes the perspective of secondary characters to narrate their past stories, thereby explaining certain character behaviors (Stein & Busse, 2009; Lanier & Schau, 2007). Through rewriting, fanfiction may also oppose specific plot points (such as stopping Riff's evil deeds in fanfiction) or alter character roles (e.g., turning Magenta into a righteous character who stops Riff's evil deeds) (Stein & Busse,

2009). This indicates that Fanfiction not only provides more substantial portrayals of secondary characters but also redefines villains more positively (Brottrager et al., 2023).

Regarding character relationships, we found that Janet and Brad frequently appear as a pair in fanfiction, consistent with their pairing in the film. However, this couple is not the most frequently occurring pair in the favorite verb type category. It suggests that fans do not show greater enthusiasm for continuing their romantic relationship in their stories. Instead, we discovered a new character pairing, Frank-Riff. Also, we observed another pairing, Janet-Frank. Although these characters shared a close relationship in the film, their relationship was not as intimate as the married couple of Janet and Brad. We also noted a new pair not seen in the film—Frank-Magenta. This pair appears in the favorite type network with high weight ten times. Such findings support Stein and Busse (2009), who stated that fanfiction often focuses on and responds to significant plot developments from the original story, either showing support or opposition by rewriting stories and adjusting character relationships.

In the subsequent analysis, we identified three pairs of characters described as having romantic relationships in fanfiction stories: Magenta-Columbia, Brad-Janet, and Magenta-Riff. Apart from Brad-Janet, who have a romantic relationship in the original film, the other two pairs are supporting characters in the film, serving to advance the plot. This finding aligns with the notion that fanfiction is often described as "filling in the gaps" of the original work, expanding possibilities not explored in the canon (Lee, 2011). As supporting characters, the relationships among Magenta, Riff, and Columbia are not depicted in the film, appearing only in a few scenes or lines of dialogue. Due to this lack of detailed portrayal, these supporting characters become entry points for fanfic authors to fill the "gaps."

Moreover, as Lanier and Schau(2007) pointed out, fanfiction tends to shift the focus of different elements in the original work. In the film, Frank's "Alternative" is the core element of the story and its focus (Grant,1991). In the corresponding fanfiction stories, we observe an expansion of character relationships beyond the elements emphasized in the canon.

From another perspective, this study has prompted reflection on the use of computational tools to abstract story texts into networks. The practice of combining network analysis and visual analysis has facilitated the implementation of a new collaborative method that integrates close reading and distant reading. However, abstracting texts into networks inevitably leads to some loss of information. This challenge stems not only from the demands of computational methods but also from the diversity of textual data.

5.3 Discussion of the Results of Character Networks Constructed Using Different Methods

In the following sections, we reflect on the information relevant to our research question that different network construction methods provide, as well as the information they may omit. We also consider how these factors impact our study, with a particular focus on critically reflecting on the methods used to construct networks.

In our discussion, we introduce the concept of relational data (Graham et al., 2016, p.183) to aid our analysis of the three methods of constructing character networks.

5.3.1 discussion about the first method.

In the first method, the relationship data we collected consists of the co-occurrence of

character names within a certain text distance and the frequency of their appearance. From the results presented, we can clearly observe which characters were used in a single story and which characters were mentioned multiple times. This allows us to infer the major and minor characters in the corresponding story. Such information helps us identify which characters from the original work have been used in fan fiction and whether any minor characters have been changed into major roles.

Although, due to the following limitations of the construction method, the information this network provides about character interactions is also limited. Firstly, since we are inferring those two characters appearing within a certain distance of each other are interacting, the co-occurrences identified by this method may not be accurate. The connections displayed in the network do not necessarily indicate substantial interactions between the characters.

Secondly, when segmenting the text windows, we only considered the length of the story. Applying a corresponding length and consistent segmentation ratio to each story may seem objective, but this approach fails to account for the development of the story's timeline and the need for scene segmentation. As the timeline progresses, character interactions may change, and different scenes involve different characters.

Moreover, during the analysis, we found that not all authors use conventional narrative structures, such as those unfolding with the progression of time. Some use poetry, diaries, or other narrative forms. This leads to situations where two characters appearing within the same text window may not actually interact in the same scene. Even if they are in the same scene, they might not engage in real interaction. They could simply be mentioned, imagined, or a scene transition might have occurred within the same text window. These inaccuracies are reflected in the network.

Despite these limitations, some scholars argue that such co-occurrence patterns can still convey meaningful co-occurrence information, as two co-occurring characters are almost always related in some way, and this relatedness is a result of integrating the entire narrative (Labatut & Bost, 2019).

5.3.2 discussion about the Second method.

In the second method of constructing character networks, we build the network based on direct conversational interactions between characters. From the results, we can gather the following information: first, characters in the story are engaged in verbal interactions and which character combinations have more frequent exchanges. Secondly, we can identify which characters play key roles in the network, such as acting as "information hubs." Using such information helps us identify character combinations with close relationships in the story and determine which characters engage in information exchanges with others.

Similarly, due to the limitations in the network construction process, the information we obtain is also limited. The extraction process in this work has demonstrated that detecting character dialogue interactions using computational methods is generally more challenging than detecting co-occurrences. Therefore, we opted for manual annotation. However, this method has the following limitations:

First, in collecting co-occurrence data, we did not distinguish between the sender and receiver of the dialogue, which means we cannot determine the direction of information flow. Since information flows from the sender to the receiver, communication is

inherently asymmetrical (Labatut & Bost, 2019, p.19), leading to the loss of valuable interpersonal relationship information. Secondly, compared to constructing character dialogue networks for texts rich in verbal interaction, such as plays, constructing dialogue networks for fan fiction in this study reflects limited aspects of character relationships (Labatut & Bost, 2019). This is because some story authors may not focus on building dialogue interactions, or the proportion of dialogue in the narrative might be small. Besides, in the manual annotation process, there was no secondary annotator to check the results, which could introduce errors due to subjective judgment. For instance, continuous dialogues in the story may not always be spoken by the same character, requiring inference about the speaker during identification. That may result in imprecision in co-occurrence detection.

In relation to the research questions, constructing a dialogue network for characters provides certain insights for answering it. First, we can observe which characters engage in dialogue within the story, with verbal interactions assumed to be co-occurrences, as two characters must appear simultaneously to converse. From this perspective, using conversation networks to assess character co-occurrence is more reliable and accurate compared to the first method based on text distance. Second, we found that the network constructed using the second method is significantly smaller than the network built using the first method. The differences help us identify which characters are excluded from dialogue and clarify the absence of specific interactions for certain characters. Since a detailed comparison involves a large amount of time, we did not conduct the process.

5.3.3 discussion about the Third method.

In the third method of constructing the network, we considered direct interactions between characters through actions. Compared to the character network constructed based on text windows, this method provides more information about different types of character relationships. Since descriptions of character interactions through actions mostly occur in narrative sections rather than dialogue, this method offers a more specific complement to identifying character interactions than focusing solely on dialogue exchanges.

Specifically, we can first identify which characters are described as interacting through actions in the story. The connection weights help us recognize the characters and combinations of characters that receive more attention from the author. Additionally, by observing the structure of connections in the network, we identified character combinations that did not appear in the previous two types of networks but appeared multiple times with high weights in the action-based network, thus enhancing our observations of changes in character relationships.

Moreover, analyzing the character network constructed based on verb types reveals that even with the same character combinations, different stories describe their interactions differently. We also noticed variations in the number of characters within specific verb-type networks, reflecting the authors' emphasis on different types of interactions. Furthermore, through this analysis, we identified romantic relationships between some specific characters in the stories.

The limitations of this method have also resulted in some information being overlooked. First, when classifying verbs, we only considered their literal meanings. Nevertheless, in the English language, the same verb can convey different meanings in different contexts, which can affect the accuracy of the network mapping. Also, we did not categorize actions in a way that aligns more closely with narrative analysis, such as actions that influence story development, indicate support, or involve physical conflict (Labatut & Bost, 2019,

p. 20). Apart from that, to simplify the data extraction process, we did not differentiate between the characters performing the actions and those receiving them.

While we considered the action categories described by verbs, the directness of the actions, and incorporated the emotional factor of affection to allocate scores to reflect interaction intensity, the visual results of the network show that merely observing the composition of characters in the network does not provide accurate information about the types of character relationships. These errors may stem from the following aspects. First, in constructing the verb corpus, we had more verbs categorized as describing everyday actions than those categorized under their respective types. This might be why most articles show higher network density under the action type. Second, there are limitations in our code design. Our code relies on punctuation to segment sentences in the story, but not all sentences in the story follow standard punctuation rules. This can lead to inaccuracies in recognition and result in extracted sentences not meeting our ideal design requirements in some stories.

In this study, we constructed character co-occurrence networks from fan fiction using three different methods based on different settings for character interaction information. These methods focus on different aspects of character co-occurrence: simple co-occurrence of character names within a certain text distance, co-occurrence of character dialogues, and co-occurrence networks based on verb classification, focusing on action interaction types and relationship formation between characters. The differences between methods provide various perspectives for observing character pairings. Future research can further refine the extraction of relational data to improve the accuracy of relationship mapping in networks. For example, for the first method, we can segment the narrative window by scenes. For the second method, we can further distinguish between information senders and receivers. For the third method, we can more finely classify the types of actions based on the emotions expressed, such as friendliness, affection, or aggression.

5.4 Limitations and Constraints

In this section, we discuss the limitations encountered in our study, particularly those related to data and the methodologies employed.

5.4.1 Limitations in Data Analysis

In this study, we were unable to directly identify how different stories organize character backgrounds, story settings, and changes in character identities (e.g., transitioning from a villain to a hero), or specific plots such as resurrection from death, solely by observing the network's node structure or calculating the network node characteristics. Such information can be identified through reading the stories. Since the definitions, collection, and visualization of character interaction information were handled differently in this research, and during the process of abstracting text content into data suitable for input and network formation, some information inevitably got lost as discussed above.

In addition, we also selected the network features that were calculated and interpreted the network and the text content based on these results. Although these calculations help us capture certain network attributes that cannot be obtained solely by reading the text or observing images, and these data form part of our argument, we must acknowledge that this data is not entirely objective. The choice of calculation methods and the interpretation of the results are influenced by personal perspectives and interpretations. As Graham et

al. (2016, p.167) pointed out, any visualization we create is imbued with the narrative and purpose we attribute to it, and our decisions on how to encode and present data significantly influence the reader's understanding of the visualization.

5.4.2 Limitations in Processing Textual Data

First, we found that the narrative style of stories affects the accuracy of data extraction. Through close reading method, we discovered that network graphs cannot fully reflect the actual character interactions in stories. We read five stories featuring dialogue verbs with high network density. In the story##1061993, the first half is composed of poetry, while the latter half involves character interactions, yet the network graph failed to accurately reflect which characters engaged in dialogue or described interactions. Similar situation was found in story##12787710, the story is narrated in the first person throughout, lacking direct dialogues but including some dialogue descriptions, such as "...talking, but I could not hear what they were saying." and "...I scream for Frank, and Riff Raff whips around towards me." These descriptions were shown in the network graph.

In analyzing story ##13917528, we found that the relationship between Riff and Magenta was reflected in the dialogue verb category of the network, with the story describing "dialogue interactions" through declarative sentences like "...he seductively whispers as he steps onto the metal elevator and presses a button." In the story ##1409260, while we could find corresponding dialogue descriptions for most characters connected in the network, the character Eddie was only mentioned in dialogue but did not actually appear, yet was incorrectly represented in the network graph, such as in the line, "'To Eddie," Brad says, and gravely the aliens raise their glasses.' From these analyses, we conclude that the narrative style of the story also affects the accuracy of relationship information extraction using computational methods.

Besides, fan fiction texts come in various forms. In our sample, stories were conveyed through poetry, lyrics, and diaries, among other forms. Such diverse text structures pose difficulties for NLP-based dialogue or character name recognition. Furthermore, an article that is too short or lacks dialogue descriptions complicates network construction. When we cannot directly print the character network through code, we have to refer back to the original text for answers.

5.4.3 Limitations in Using Data Processing Tools

Firstly, the method of character recognition mainly relies on identifying character names in the text. Despite considering multiple variations of character names in the text, during the experiment, we could not accurately recognize character references. If an author's writing style involves extensive use of pronouns, without context, identifying the referenced characters becomes challenging, potentially leading to underrepresented data results.

In addition, due to my limited computing background, some inaccuracies may exist in the code implementation. To minimize these inaccuracies as much as possible, I repeatedly modified the code during the experiment and consulted relevant materials to ensure that the libraries and methods in use aligned with the expected functions, adding detailed explanations in the code to clarify the logic and purpose of each step. Nonetheless, these limitations are objectively existed.

5.5 Further Research

Although our analysis in this study addressed the romantic relationships between characters, we did not consider the hierarchical relationships between characters. We cannot deduce the positions assigned to characters in the story based solely on the amount of dialogue interaction or specific types of action interactions. Sometimes hierarchical relationships, such as sibling relationships, are explicitly mentioned in the narrative, but some authors imply these relationships through the characters' terms of address in dialogue. Therefore, in future research, scholars should consider including hierarchical or power relationships when discussing character relationships.

In future research, we can use different computational methods to study the reconstruction of character relationships and interactions in fan fiction. We can conduct sentiment analysis on the descriptions of character relationships in the stories, observing whether the authors use positive or negative terminology when describing specific character pairings. Additionally, we can perform topic modeling on fan fiction texts, integrating identified character relationships within the stories, and analyze the plot elements that these relationships might participate in.

Finally, the criteria for selecting fan fiction stories and the methods for identifying character relationships vary depending on the specific research objectives. In future research, researchers can opt for comparative studies of fan fiction from different regions or languages or construct character networks using other types of data. These thematic variations could offer us different results and perspectives.

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